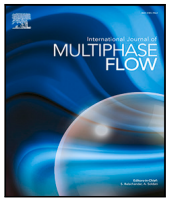
International Journal of Multiphase Flow 172 (2024) 104715



Contents lists available at ScienceDirect

International Journal of Multiphase Flow

journal homepage: www.elsevier.com/locate/ijmulflow



Leadership Contribution



Recent advances in well-posed Eulerian models for polydisperse multiphase flows

Rodney O. Fox *

Department of Chemical and Biological Engineering, Iowa State University, Ames, IA 50011-1098, United States of America
Laboratoire EM2C, CNRS, CentraleSupélec, Université Paris-Saclay, 3 rue Joliot-Curie, 91192 Gif-sur-Yvette, France
Center for Multiphase Flow Research and Education (CoMFRE), College of Engineering, Iowa State University, United States of America

ARTICLE INFO

Keywords:
Polydisperse multiphase flow
Kinetic-based model
Quadrature-based moment methods
Well-posed Eulerian models

ABSTRACT

The current state-of-the-art for computational modeling of polydisperse multiphase flows is reviewed and future research directions are discussed. Physics-based computational models at three distinct levels of abstraction: the microscale, mesoscale and macroscale; are discussed and compared. Special emphasis is placed on the relationship between the models at different scales and on how information from the finer scales is used to provide closures at the coarser scales. For disperse multiphase flows, it is argued that the passage from a direct-numerical simulation at the microscale to the kinetic description at the mesoscale is the crucial step for ensuring the validity of the macroscale model. In particular, the passage from the microscale to the mesoscale requires physics-based closures, while the passage from the mesoscale to the macroscale requires mathematical closures. The choices made in the physical and mathematical closures of the spatial fluxes and coupling terms will determine whether the Eulerian model is well-posed. In addition, the use of quadrature-based moment methods for polydisperse particles is presented as an efficient macroscale closure when dealing with a distribution of particle sizes. Examples of monodisperse and polydisperse multiphase flows are provided for cases where the fluid phase is compressible (high speed) and incompressible (low speed).

1. Introduction

Polydisperse multiphase flows exist in many different forms and occur in numerous applications (Balachandar and Eaton, 2010; Guazzelli and Hinch, 2011; Deike, 2022; Brandt and Coletti, 2022; Morris, 2020; Capecelatro and Wagner, 2024; Capecelatro, 2022; Risso, 2018; Prosperetti, 2017; Voth and Soldati, 2017; Dufek, 2016). For the purposes of this work, a *disperse* multiphase flow consists of a disperse phase (gas, liquid or solid) and a continuous phase (gas or liquid). The individual elements making up the disperse phase are referred to as *particles* (e.g., bubbles, droplets, solid spheres, etc.). Furthermore, if the disperse phase is made up of particles with identical physical properties (volume, shape, density, temperature, composition, surface tension, etc.), then the flow is *monodisperse*. Otherwise, if some properties vary from particle to particle, then the flow is *polydisperse*. In either case, the particle velocities are allowed to be different for each particle, varying inside the flow due to surface stresses in the continuous phase and body forces acting on the particles. Other important subcategories for the particles are *deformable* and *non-deformable* and, for the continuous phase, *Newtonian* and *non-Newtonian* fluids. For clarity, in this work, the main focus is on monodisperse and non-deformable two-phase flows with Newtonian fluids, but the extension to polydisperse particles is also discussed (Fox et al., 2024). In any case, the modeling approaches are also applicable for more complex cases where, for example, the particles' shapes change due to hydrodynamic forces (as in the example in Fig. 1). Such polydisperse cases require additional variables (beyond mass and velocity) to describe the state of a particle (Marchisio and Fox, 2013; Essadki et al., 2019; Fox et al., 2024). This extension is briefly described for polydisperse gas–particle flows in Section 4.2.

The primary objective of this work is to provide a high-level overview of recent advances in formulating Eulerian models for polydisperse multiphase flow. In the literature, this type of model is referred to as a two-fluid (monodisperse) or a multi-fluid (polydisperse) model. Mathematically, such models consist of (at a minimum) continuum balances for the mass and momentum of the two phases (disperse, continuous) that are coupled together through various interfacial force terms. As shown in Fig. 2, there are at least two methods commonly used to derive these balance equations for polydisperse multiphase flows. However, one very important mathematical property needed for a well-posed two-fluid model is that the system of (inviscid) balance equations be *hyperbolic* (Lhuillier et al., 2013). In practice, the main

* Correspondence to: Department of Chemical and Biological Engineering, Iowa State University, Ames, IA 50011-1098, United States of America.
E-mail address: rofox@iastate.edu.

https://doi.org/10.1016/j.ijmultiphaseflow.2023.104715
Received 28 October 2023; Received in revised form 10 December 2023; Accepted 22 December 2023
Available online 27 December 2023
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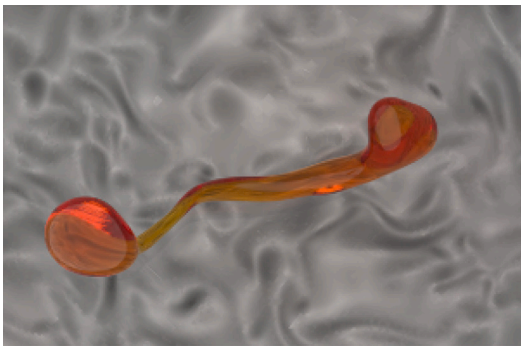


Fig. 1. An oil droplet in water subject to homogeneous isotropic turbulence. The droplet shape and eventual breakage into smaller drops depends on the Reynolds and Weber numbers, as well as the viscosity ratio between the two phases (Farsoiya et al., 2023). For this polydisperse flow, the particles are deformable so that both the mass and surface area are needed in the kinetic-based approach. Image courtesy of Palas Kumar Farsoiya and Luc Deike.

cause of ill-posed multiphase flow equations is the Archimedes' force exerted by the continuous phase on the disperse phase, particularly for flows where the fluid material density ρ_f is larger than the particle density ρ_p (e.g., bubbly flows). As discussed in detail below, one of the most important recent advances is a globally hyperbolic Eulerian model based on a kinetic theory accounting for particle–particle interactions transmitted by the continuous fluid phase (Fox et al., 2020), which are referred to as the particle–fluid–particle (pfp) (Wang et al., 2021) or hydrodynamic (Spelt and Sangani, 1998) stresses.

It must be emphasized that kinetic-based models can only be applied to the disperse phase when it is made up of discrete particles, for example, with bubbles (Spelt and Sangani, 1998), droplets (Essadki et al., 2019), or solids (Gidaspow, 1994; Simonin et al., 1993). Thus, the Eulerian models described in this work are a subset of the two-fluid models used in the literature (Drew and Passman, 1999). For example, 1-D two-fluid models developed for (turbulent and/or segregated) gas–liquid flow in pipelines do not fall into the subset of kinetic-based Eulerian models. Another important distinction is that the kinetic-based models do not employ Reynolds averaging as is done in single-phase turbulence models. Thus, at high-enough Reynolds numbers, the flow predicted by a kinetic-based model will be turbulent at length scales much larger than the particle size. Therefore, in the limit where the volume fraction of the disperse phase α_p is very small, the continuous phase will be governed by the Navier–Stokes (or Euler) equation. In this sense, its numerical solution is a direct-numerical simulation (DNS) of the kinetic-based model. Nonetheless, a kinetic-based two-fluid model can also be Reynolds averaged or spatially filtered to develop a multiphase turbulence model (Fox, 2014, 2021; Ozel et al., 2013). However, this step is completely separate from the kinetic-based modeling discussed in this work.

As shown in Fig. 2, particle-resolved DNS (PR-DNS) is referred to as the *microscale* solution (Tenneti and Subramaniam, 2014; Elghobashi, 2019; Maxey, 2017), DNS of the kinetic-based model as the *mesoscale* solution, and DNS of the hyperbolic conservation equations derived from the kinetic-based model (i.e., the two-fluid model) as the *macroscale* solution. At each level of description, the number of conserved variables will be smaller as details at the finer level appear as constitutive equations at the coarser level. For example, the continuous-phase stresses at any point on a particle surface are known at the microscale, but are represented as a fluid–particle drag force for each particle at the mesoscale (i.e., averaging the point-wise stresses over the particle's surface). At the macroscale, only the fluid–particle drag force for the *average* particle velocity appears in the momentum balances. The modeling challenge is then to make the descriptions at the different levels be consistent with each other, especially for flows wherein the disperse-phase volume fraction is large enough to have strong particle–particle interactions.

The flow physics of polydisperse multiphase flows exhibit a strong dependence on the material density ratio $\mathcal{Z} = \rho_f/\rho_p$ and the disperse-phase volume fraction α_p. Generally speaking, it is more difficult to model flows with $\mathcal{Z} \gg 1$ (e.g., bubbly flows Spelt and Sangani, 1998) than with $\mathcal{Z} \ll 1$ (e.g., gas–solid flows Gidaspow, 1994). This is primarily because with larger $\mathcal{Z}$, the role of the fluid in determining the spatial fluxes and interfacial stresses is more important and multifaceted. For example, in gas–solid flows the principal coupling term is the fluid drag force on the particles, and particle-phase stresses arise due to particle–particle collisions. For these flows the added mass and lift forces (which scale with ρ_f) can usually be neglected. On the other hand, for bubbly flows the added-mass and lift forces are much larger than direct particle–particle collisions. Moreover, the pfp stresses (which scale with ρ_f Bulthuis et al., 1995) must be taken into account when $\mathcal{Z} > 0.1$ (e.g., solid–liquid sedimentation).

The mass loading, defined by $\varphi = \rho_p\alpha_p/(\rho_f\alpha_f)$ with continuous-phase volume fraction $\alpha_f = 1 - \alpha_p$, is also an important parameter that determines the multiphase flow regime. For example, with $\varphi \ll 1$, the continuous-phase momentum balance is weakly dependent on the disperse phase, while for $\varphi \geq 1$ most of the flow momentum is present in the disperse phase. Finally, another important quantity is the phasic slip velocity $\mathbf{u}_{pf} = \mathbf{u}_p - \mathbf{u}_f$ where $\mathbf{u}_p$ and $\mathbf{u}_f$ are the average velocities in the particle and fluid phases, respectively. For very small particles (i.e., Stokes number St near zero), the slip velocity can be neglected so that only the continuous-phase momentum balance must be solved, in which case the model is almost always well-posed. However, a very small, but finite, slip velocity leads to ill-posed models when $\mathcal{Z} \gg 1$. In the kinetic-based approach, all important terms observed in PR-DNS can, in principle, be included so that the macroscale model will be applicable and well-posed for arbitrary values of $\mathbf{u}_{pf}$, φ, $\mathcal{Z}$ and α_p.

The remainder of this work is organized as follows. In Section 2, the multi-scale modeling approach for polydisperse flows is discussed, as well as the important physical parameters that must be accounted for at the three different scales (microscale, mesoscale, macroscale). In Section 3, examples of the models developed at each of the scales are provided in order to familiarize the reader with the modeling challenges encountered at each scale and with how information from finer scales is used at the coarser scales. The two most common types of mesoscale models (Euler–Lagrange, kinetic equation) are also compared. In Section 4, a specific example of kinetic-based modeling of gas–particle flow is presented in sufficient detail to introduce the reader to this critical modeling step. Then, in Section 5, this kinetic model is employed to find the unclosed macroscale model equations, which are subsequently closed using quadrature-based moment methods (QBMM). Finally, in Section 6, example applications of the macroscale models are provided to demonstrate the wide range of physics that can be captured using well-posed Eulerian models. Conclusion are drawn in Section 7.

2. Computational modeling of polydisperse multiphase flow

Before going into details with specific examples, in this section, a bit more information about the modeling approach used for disperse multiphase flows is presented. In particular, the physical distinctions between the models used at the different scales are clearly discussed. Note that the term 'particle' is used to denote the property of a single particle, and 'particle-phase' to denote the average property of all particles. For example, at the microscale, the particle velocity $\mathbf{U}(t)$ and location $\mathbf{X}(t)$ are known, and, at the macroscale, the particle-phase velocity $\mathbf{u}_p(t,\mathbf{x})$ is known. In between at the mesoscale, the kinetic equation uses the number density function (NDF) $f(t,\mathbf{x},\mathbf{u})$, just as is done in the kinetic theory of molecular gases with the Boltzmann equation (Sirovich, 1962). As with time t and space $\mathbf{x}$, in the NDF the velocity $\mathbf{u}$ is an independent variable in three dimensions such that

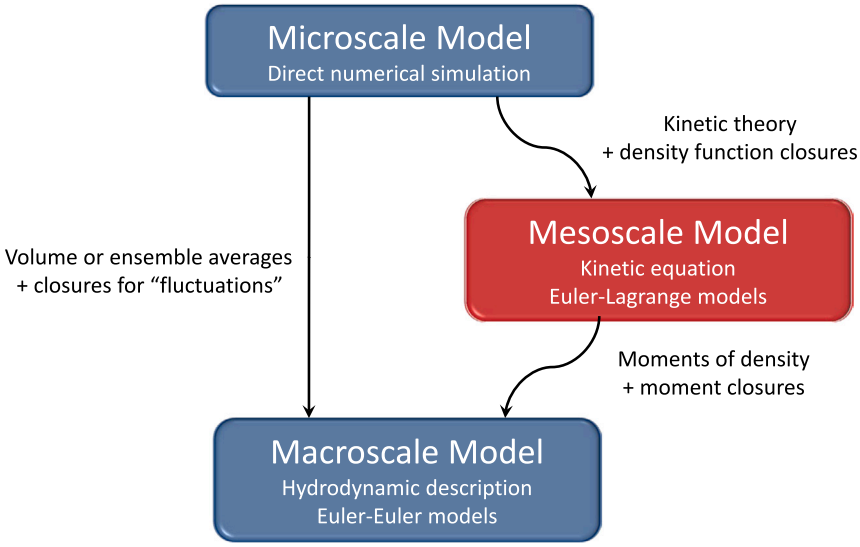


Fig. 2. Polydisperse modeling framework for the kinetic-based approach involving three levels: microscale, mesoscale, and macroscale (Marchisio and Fox, 2013). The kinetic model is formulated at the mesoscale, using data from the microscale, and employed to derive a well-posed macroscale model.

f involves seven independent variables in a seven-dimensional (7-D) space.

For polydisperse flows (Marchisio and Fox, 2013; Fox et al., 2024), more independent variables $\boldsymbol{\xi}$ are needed in f to describe the particle properties besides velocity. For example, the GeoMOM approach for droplets uses properties related to the geometry of the particle surface (Essadki et al., 2019). Likewise, at the microscale, individual particles will have properties $\boldsymbol{\xi}(t)$. Ultimately, at the macroscale, the mathematical description is reduced to only a few moments of $f(t, \mathbf{x}, \mathbf{u}, \boldsymbol{\xi})$, i.e., the particle-phase properties, that depend on $(t, \mathbf{x})$ and can be solved for using standard finite-volume methods on 3-D unstructured meshes (Passalacqua et al., 2018). However, depending on the system, there is no *a priori* guarantee that the reduction from the microscale to the macroscale will be successful. Indeed, it may be possible to derive a predictive mesoscale model that cannot be reduced to a tractable macroscale model. A well-known example where this occurs is rarefied gases flows with large Knudsen number.

2.1. Modeling framework

Returning to Fig. 2, the modeling framework for polydisperse multiphase flows can be divided into three levels. At the finest level, the microscale model resolves all of the physics present in the flow, and its solution is treated as the *ground truth* to which all higher-level models can be validated. Traditionally, two-fluid models have been derived from the microscale model by applying volume or ensemble averages (Drew and Passman, 1999). Due to non-linearity, this process leads to unclosed terms for the spatial fluxes and surface stresses on the interface between the phases. The closures for such terms involve complex physics and are generally difficult to formulate for even simple cases (e.g., with monodisperse particles). Moreover, *ad hoc* closures do not guarantee that the resulting macroscale model will be well-posed. Fortunately, it is usually possible to develop a kinetic-based mesoscale model for the disperse phase.

As will be discussed below, there are different types of mesoscale models, falling into two broad categories: Euler–Lagrange and kinetic models. For the former, the continuous fluid is handled in an Eulerian framework while each particle in the disperse phase in treated with a Lagrangian approach (Capecelatro and Desjardins, 2013). The coupling between the continuous and disperse phases in done using correlations, which preferably have been validated with microscale DNS (Tenneti and Subramaniam, 2014). Alternatively, the kinetic approach uses a distribution function for the particle number concentration (or NDF) that depends on space, time, and the variables describing the state of a particle in the Lagrangian approach (Marchisio and Fox, 2013). For example, for monodisperse particles the usual state variable is the particle velocity, while for polydisperse particles the particle mass also appears as a state variable.

While the mesoscale model provides much more information than the macroscale model, its computational cost is also much higher. However, because Euler–Lagrange and kinetic models have an explicit representation of the particle microscale physics, the physical closures used at the mesoscale are simpler to formulate than those needed in volume/ensemble averaging at the macroscale. At the mesoscale, the Euler–Lagrange and kinetic models are complementary in the sense that the former can be used when the number of particles is not too large, while the latter works best when the number of elements is very large. However, the real advantage of working with the kinetic model is that it offers a convenient route towards the macroscale model.

In addition, because the kinetic model retains as much microscale physics as possible, the resulting macroscale model is likely to be well-posed. Finally, it is important to note that the path from the mesoscale to macroscale models relies on mathematical (and not physical) closures. In other words, the mesoscale model can, in principle, be solved exactly and this solution could then be used to find the macroscale variables. However, by using a mathematical approximation for the mesoscale solution, we can derive a *closed* macroscale model. In contrast, going from the microscale to the mesoscale model requires physical simplifications, such as a drag model or a collision model, that do not directly follow from the microscale equations. This difference allows for a clear separation between physical modeling and mathematical approximations that are mixed together in volume/ensemble averaging. In summary, because the mesoscale model requires *mechanistic* models for particle–particle and particle–fluid interactions, it incorporates more microscale physics that will eventually end up in the macroscale model (Marchisio and Fox, 2013).

2.2. Modeling objectives

The purpose of the kinetic-based Eulerian model is to reproduce the important flow dynamics present at the microscale. At a minimum, we want the macroscale model to be well-posed and to be able to treat polydisperse particles interacting with a high-speed or low-speed continuous fluid phase. Here, high speed refers to compressible flows,

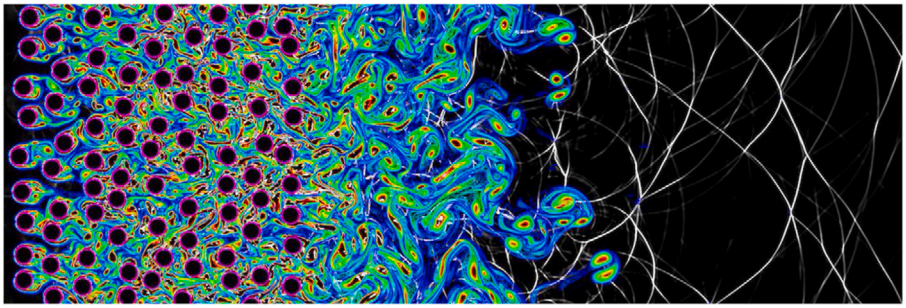


Fig. 3. Pseudo-turbulent kinetic energy (PTKE) generated by a shock wave traversing a curtain of solid particles (Shallcross et al., 2020). PTKE is produced inside the curtain and transported downstream by the fluid flow. Results shown here are from a direct-numerical simulation of the microscale model with fixed particle positions. The fluid-phase PTKE must be modeled at the mesoscale and the macroscale, and coupled to the model for the particle-phase granular temperature. Image courtesy of Gregory Shallcross and Jesse Capecelatro.

including supersonic cases, whereas low speed refers to incompressible or low-Mach-number flows. As mentioned earlier, polydisperse in this work refers to particles with different masses (i.e., a particle size distribution) with possibly different temperatures. Another important objective for the macroscale model is to have the ability to handle very dilute ($\alpha_p \ll 0.1$) to very dense ($\alpha_p > 0.1$) regions in the same flow. Here, dilute/dense refer to the volume fraction of the disperse phase. For example, for very dilute flows particle–particle collisions can be negligible, which is not the case for dense flows. Furthermore, the kinetic model must be valid for all possible values of the mass loading φ. Thus, the kinetic-based model must include all of the relevant physics present in these different regimes.

For polydisperse particles, the velocity of a particle can be very much dependent on its mass (Marchisio and Fox, 2013). In other words, the flow will generate a strong correlation between particle velocity and particle mass, leading to mesoscale particle velocities that depend on the mass. At the kinetic level, such correlations are easily captured because the fluid forces acting on individual particles (e.g., fluid drag) will depend on the particle's size. However, at the macroscale, numerous unclosed terms appear in the particle-phase momentum balance that depend on the size distribution. Thus, a consistent and numerically robust closure procedure is required to ensure that the polydisperse Eulerian model remains well-posed as the particle size distribution evolves spatially due to the mass-dependent velocity. This is especially true when the size distribution is very board, spanning one or two orders of length scales (e.g., microns to millimeters) in many applications.

In general, finite-sized particles can have significantly different velocities than the surrounding fluid and, therefore, generate a wake. In many cases, the wake is turbulent and, in denser flows, the wakes of nearby particles overlap and interact. This is shown in Fig. 3 for a high-speed flow passing through a curtain of heavy particles from a PR-DNS. In the literature, the turbulent flow generated by the wakes is often referred to as *pseudoturbulence* or pseudoturbulent kinetic energy (PTKE) (Risso, 2018). It is important to note that the characteristic length scale of PTKE is on the order of the particle diameter d_p. Nevertheless, as seen in Fig. 3, the PTKE is convected downstream from the particle curtain where the characteristic length scale evolves as observed in grid turbulence.

At the mesoscale and macroscale, the PTKE in the fluid phase must be modeled (Shallcross et al., 2020). Moreover, depending on the particle density relative to the fluid density (i.e., on $\mathcal{Z}$), the PTKE in the fluid will accelerate the particles, giving them a 'random' motion relative to their mean velocity (Tavanashad et al., 2019). In the literature, the kinetic energy in the particle random motion is referred to as the *granular temperature* (Gidaspow, 1994) and denoted here by Θ_p. In the kinetic-based model for the particles, the fluctuating velocity is included so that an acceleration source term proportional to the PTKE must appear in the kinetic equation (Fox et al., 2020). The same term with opposite sign will appear in the mesoscale PTKE balance. It is important to understand that PTKE is very different from multiphase turbulent kinetic energy, which occurs at much larger scales and is resolved by meso/macroscale DNS. In many cases, multiphase turbulence is generated by buoyancy and results is large-scale variations in the disperse-phase volume fraction (Fox, 2021).

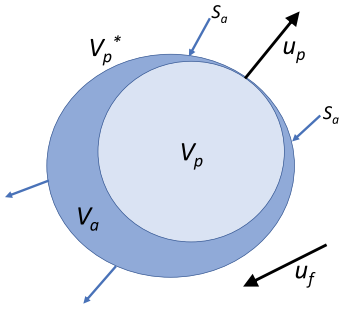


Fig. 4. Representation of the added volume V_a around a particle with volume V_p (Fox et al., 2020). At the mesoscale, $V_p^\star = V_p + V_a$ is the total volume moving with velocity $\mathbf{u}_p$, and the surrounding fluid moves with velocity $\mathbf{u}_f$. The wake adapts to its local surroundings by incorporating fluid at the rate S_a, i.e., $S_a = 0$ when V_a does not change. Note that mass flow in and out of the wake with constant V_a would be treated as mass exchange with the surrounding fluid, and is neglected here for simplicity.

When the fluid density is not small compared to the particle density (i.e., $\mathcal{Z} > 0.01$), the mass of the fluid wake surrounding a particle will not be negligible. This *added mass* can be represented as an *added volume* V_a shown in Fig. 4, which surrounds the particle volume V_p with mass $m_p = \rho_p V_p$ (Biesheuvel and Spoelstra, 1989; Cook and Harlow, 1984; Spelt and Sangani, 1998; Fox et al., 2020; Boniou et al., 2023). In the kinetic-based model, the total volume $V_p + V_a$ with mass $m_p^\star = \rho_e V_p^\star = \rho_p V_p + \rho_f V_a = \beta(t,\mathbf{x}) m_p$ has velocity $\mathbf{u}_p$, whereas the surrounding fluid has velocity $\mathbf{u}_f$. Here, $\beta(t,\mathbf{x}) = 1 + c_m \mathcal{Z} \alpha_f \geq 1$ is the local added-mass factor where $c_m = 0.5$ for isolated spheres ($\alpha_f = 1$). For $\alpha_p > 0$, the correlation

$$c_m = \frac{1 + 2\alpha_p}{2\alpha_f}$$

can be used (Zuber, 1964; Sangani et al., 1991; Spelt and Sangani, 1998) when $\alpha_p < 0.1$. For larger α_p, the constant $c_m = 2/3$ can be used. Because the size of the wake will depend on the volume fraction of the particles, mass exchange at the rate S_a occurs from the surrounding fluid to the wake. As shown elsewhere (Cook and Harlow, 1984; Fox et al., 2020), the particle-phase momentum balance written for $m_p^\star \mathbf{u}_p$ is equivalent to the virtual-mass force in the momentum balance for $m_p \mathbf{u}_p$.

In summary, for the mesoscale description based on a kinetic equation for the particles, the (monodisperse) particle volume is denoted by $V_p^\star$, and its mass as $m_p^\star$. Note that this implies that the usual fluid-phase momentum in the two-fluid model $\rho_f \alpha_f \mathbf{v}_f$ is related to $\mathbf{u}_f$ by $\rho_f \alpha_f \mathbf{v}_f =$

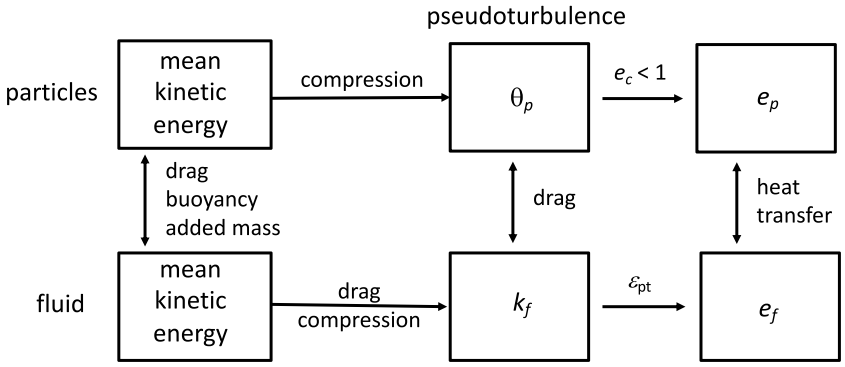


Fig. 5. Energy diagram for fluid–particle flows (Capecelatro and Wagner, 2024). For high-speed flows, it is essential to account for the energy production and exchange mechanisms in order to predict accurately the gas pressure and thus the speed of sound. For each phase, the macroscale model includes a balance for the total energy and for the internal energy. For low-speed flows with constant ρ_f, balances for pseudoturbulence (i.e., Θ_p and k_f) and internal energy are used. The thermodynamic temperatures T_p and T_f are found from the internal energy in each phase, e_p and e_f. In this work, for simplicity, heat transfer is neglected and the internal energy of the particle phase e_p is constant.

$\rho_f(\alpha_f \mathbf{u}_f + \alpha_a \mathbf{u}_p)$ where α_a is the added-mass volume fraction. Finally, for convenience, the volume fractions $\alpha_p^\star = \alpha_p + \alpha_a$ and $\alpha_f^\star = \alpha_f - \alpha_a$ are defined with the property $\alpha_p^\star + \alpha_f^\star = 1$. In this work, ρ_p is assumed to be constant, while ρ_f varies depending on the fluid pressure and temperature in high-speed flows and is (nearly) constant in low-speed flows.

For high-speed flows, the equation of state for the fluid pressure requires the fluid temperature. In a disperse multiphase flow, heat transfer between the particles and continuous fluid can have a significant impact on the fluid-phase speed of sound (Houim and Oran, 2016; Boniou and Fox, 2023). Thus, it is necessary to model the internal energy of each particle in the kinetic model, as well as the total energy of the fluid, which are coupled through heat-transfer terms (Boniou et al., 2023). In the following, the fluid-phase total energy is denoted by $E_f = \frac{1}{2}u_f^2 + k_f + e_f$ and the particle-phase total energy by $E_p = \frac{1}{2}u_p^2 + \frac{3}{2}\Theta_p + e_p$. The first term on the right-hand side is the mean kinetic energy, and the third term is the internal energy. The total kinetic energies are $K_f = \frac{1}{2}u_f^2 + k_f$ and $K_p = \frac{1}{2}u_p^2 + \frac{3}{2}\Theta_p$ where k_f represents PTKE and Θ_p is the granular temperature.

The relationship between the different forms of energy is illustrated in Fig. 5. In this diagram, T_p and T_f are the thermodynamic temperatures found from e_p and e_f, respectively. If the heat capacity is constant, then the temperature is linearly proportional to the internal energy. The symbol e_c is the coefficient of restitution for particle–particle collisions. When $e_c < 1$, Θ_p decreases during each collision and the lost energy heats the particles. Similarly, ε_{pt} represents viscous dissipation of PTKE, which heats the fluid. For the particles, 'compression' refers to regions where α_p increases, causing Θ_p to increase. Likewise, for the fluid, compression increases ρ_f causing k_f (and ultimately T_f) to increase.

2.3. Physics of polydisperse multiphase flows

As mentioned earlier, the flow physics depends strongly on the density ratio $\mathcal{Z}$. In practice, the density ratio is fixed by the nature of the fluid (i.e., gas or liquid) and particles (e.g., bubble or solid). In any case, $\mathcal{Z} = 1$ corresponds to buoyancy-neutral flows, for which the two-fluid model should reduce to a one-fluid model resembling the Navier–Stokes equation (i.e., with an effective viscosity depending on the particle volume fraction).[1] Otherwise, the coupling between the particle- and fluid-phase momentum balances depends on the value of $\mathcal{Z}$. In the context of macroscale models for non-deformable particles, the flow physics can be summarized as follows. In Section 3 examples of gravity-driven flows with different density ratios are provided.

[1] Even with $\mathcal{Z} = 1$, it is possible to have particles with a different velocity than the fluid. For example, magnetic particles in a magnetic field can be accelerated differently than the fluid. In such cases, a momentum balance may be needed for the particle phase.

$\mathcal{Z} \gg 1$. For gas bubbles in a liquid, the added mass and buoyancy will be strong, leading to a relatively large slip velocity (Spelt and Sangani, 1998; Risso, 2018). This, in turn, results in a large particle Reynolds number and a strong drag force. Because they scale with ρ_f and the square of the slip velocity, the pfp stresses are also very important in bubbly flows, as can be the lift force. Furthermore, larger bubbles can deform significantly, which must be accounted for in the various coupling terms. Due to the strong buoyancy force, the two-fluid model without the pfp stresses is ill-posed (Lhuillier et al., 2013) (unless *ad hoc* forces are added). In many cases, bubble coalescence and breakage must be included in the kinetic model. The coupling with the fluid-phase momentum balance is mainly determined by α_p since the disperse phase has negligible momentum.

$\mathcal{Z} < 1$. For solid particles in a liquid, the most important coupling terms in the macroscale model are the fluid drag force and the pfp stresses, followed by added mass, buoyancy and lift. Here again the two-fluid model without the pfp stresses is ill-posed (Lhuillier et al., 2013). For larger α_p, inelastic particle–particle collisions lower Θ_p so that the particle-phase pressure is relatively small. However, due to the coupling with PTKE, Θ_p does not decay to zero as long as the slip velocity is present. Finally, at large α_p, frictional contacts between particles lead to a large particle-phase frictional pressure and viscosity. In many cases, the particles do not remain uniformly distributed in space. Instead, kinematic waves or 'clusters' form wherein α_p is higher/lower than the local average volume fraction (Shajahan and Breugem, 2023). This implies that the two-fluid model will not have steady-state or simple 1-D solutions.

$\mathcal{Z} \ll 1$. For solid particles or liquid droplets in a gas, the most important coupling term in the macroscale model is the fluid drag force. All other terms (including buoyancy) scale with $\mathcal{Z}$, and are therefore small in low-speed flows. Nonetheless, in high-speed flows the pfp stresses can be very large due to the large slip velocity (Boniou and Fox, 2023). Although the particle pressure due to particle–particle collisions can be significant, it is usually not enough to remove large variations in α_p (referred to as 'clusters' or 'swarms' in the literature). The values of the variables for which the two-fluid model is ill-posed are fairly rare (Fox et al., 2020) (or nonexistent if the small buoyancy force is set to zero). However, due to the large particle-phase Mach number, the flow is highly unsteady and exhibits steep shocks in α_p that are difficult to resolve in numerical simulations (Serna and Marquina, 2005).

Role of α_p. In addition to $\mathcal{Z}$, the value of α_p can be used to designate different flow regimes (Boniou and Fox, 2023). In this work, macroscale models valid when the particle-phase velocity moments are nearly Maxwellian (in the sense of kinetic theory) are considered. For such cases, it suffices to solve for the mass, momentum and total energy of the particle phase. This will be the case when α_p and Θ_p are large

enough to generate significant particle–particle collisions. Nonetheless, by including higher-order velocity moments, it is possible to treat weakly collisional flows (Vié et al., 2015; Fox and Laurent, 2022). In any case, the mesoscale kinetic equation is valid for collisional as well as non-collisional flows. Thus, it is only a question of augmenting the macroscale model to account for such behavior. Finally, note that the mass loading φ will be large when $\mathcal{Z} \ll 1$, in which case the mixture momentum is dominated by the particle phase.

3. Examples of models at different scales

In this section, some specific examples of models used at the three scales shown in Fig. 2 are provided. Since it is the essential starting point for developing well-posed macroscale models, the mesoscale model based on kinetic theory is described in more detail than the microscale and macroscale models. Furthermore, the numerical methods needed to solve each type of model are briefly mentioned. Most importantly, the key conceptual points to understand from this section are (1) how the unclosed terms are related between the different scales, and (2) how their closure determines whether the macroscale model is well-posed.

3.1. Microscale models

In the context of polydisperse multiphase flows, the microscale model contains all of the relevant flow physics. In this sense, its solution is a particle-resolved, direct-numerical simulation (PR-DNS) (Tenneti and Subramaniam, 2014). For example, isothermal particles in a Newtonian fluid require the resolution of the fluid velocity field, the interfacial fluid stresses on the particles' surfaces, and Newton's equations for the linear and angular momentum of each particle. In practice, care must be taken to handle correctly particle–particle collisions (Maxey, 2017). When the particles are deformable (e.g., gas bubbles or droplets), one must also resolve the velocity field inside the particles and handle the interfacial stresses, etc. In short, a true PR-DNS requires specialized numerical algorithms that ensure that all of the physics is captured accurately.

The obvious advantage of PR-DNS is that it provides all data required to understand the complex flow physics. Such data are usually impossible to measure experimentally for multiphase flows, but are vital for validation of mesoscale closures. There are also obvious limitations such as the high computational cost, the large data sets that must be manipulated to find quantities relevant to specific applications, and the difficulty of applying PR-DNS with complex geometries, all of which restricts PR-DNS to relatively small-scale systems. It is precisely for this reason that microscale models are used to develop mesoscale models applicable to large-scale systems using the same or less computational resources. For example, the microscale model can be solved for a canonical flow (such as statistically homogeneous flow in a periodic domain) that represents a sub-domain in a larger-scale flow.

As an example of PR-DNS for fluid–solid flow, Fig. 6 shows a snapshot of bi-disperse particles (i.e., two diameters, but same density) in a periodic domain (Tenneti and Subramaniam, 2014). A body force (e.g., gravity) keeps the time-averaged particle velocity constant and the time-averaged fluid velocity is fixed by a mean pressure gradient. The time-averaged slip velocity between the two phases is then constant. The resulting particle Reynolds number Re_p is large enough for wakes to form behind individual particles. The instantaneous velocity of each particle exhibits fluctuations (related to granular temperature) that are particle-size dependent. In general, the large particles have smaller fluctuations than the small particles, which is expected due to the ratio of their masses. The fluid velocity exhibits high-speed streaks between particles, as well as low-speed regions in the particle wakes. At the mesoscale, these fluid-velocity fluctuations are treated as PTKE. Because the data for velocity and pressure are available at every spatial location and at every time step, the fluid stresses on the surface of each particle are known in a microscale simulation. In contrast, at the mesoscale such detailed information is not available. Thus, it is necessary to develop and validate a fluid-drag model to predict the average fluid stresses at the mesoscale. For low-speed flows, such a correlation will depend on Re_p and the volume fraction α_p (Tavanashad et al., 2021). Note also that the microscale model provides the fluctuating fluid stresses due to PTKE and the granular temperature of the particles of each size. This information can be used to validate mesoscale models for the pseudoturbulence (see Fig. 5).

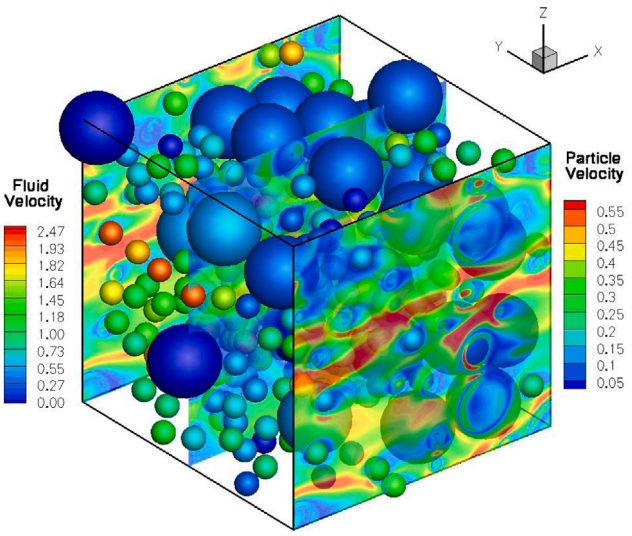


Fig. 6. PR-DNS of spatially homogeneous, bi-disperse, fluid–particle flow in a periodic domain (Tenneti and Subramaniam, 2014). The magnitudes of the particle velocity are shown in the color bar on the right, while the magnitude of the fluid velocity is shown in the color bar on the left. At the mesoscale and macroscale, these velocity fluctuations are modeled as PTKE and granular temperature. Image courtesy of Mohammad Mehrabadi, Shankar Subramaniam, and Sudheer Tenneti. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In high-speed gas–particle flows, the gas-phase Mach number Ma can be large enough to create shocks around the particles (Khalloufi and Capecelatro, 2023). This is shown in Fig. 7 taken from a PR-DNS of compressible flow around fixed particles. In the fluid-velocity fields, the wakes between particles and the high-speed streaks are clearly visible and increase in magnitude with increasing Ma. In comparison to low-speed flow, the pressure fluctuations in Fig. 7 at $Ma = 1$ are especially large and located upstream of each particle. At the mesoscale, these observations imply that the fluid-drag correlation and the model for PTKE must also depend on the gas-phase Mach number (Osnes et al., 2023).

As a final example of microscale modeling, consider monodisperse solid spheres settling in water under gravity as shown in Fig. 8 (Shajahan and Breugem, 2020, 2023). For this case, the density ratio is $\mathcal{Z} = 2/3$ and the volume fraction is $\alpha_p = 0.3$. In the simulations, the particles reach a slip velocity that depends on Re_p, α_p and the Galileo number Ga. Because of particle–particle interactions through the fluid ('hindrance effects'), this slip velocity decreases with increasing α_p such that denser suspensions settle much more slowly than dilute ones. Furthermore, particle–particle interactions in dense suspensions lead to kinematic waves (Shajahan and Breugem, 2023). The velocity of the kinematic waves depend on α_p through the drag correlation, while their frequency depends are the magnitude of particle–particle interactions through the fluid. In the mesoscale model, the model for the pfp stresses plays the role of the fully resolved particle–particle interactions in the microscale simulations, but the drag correlation fixes the slip velocity (i.e., the kinematic wave speed). Thus, it should be possible to validate mesoscale models for the pfp stresses by comparing the

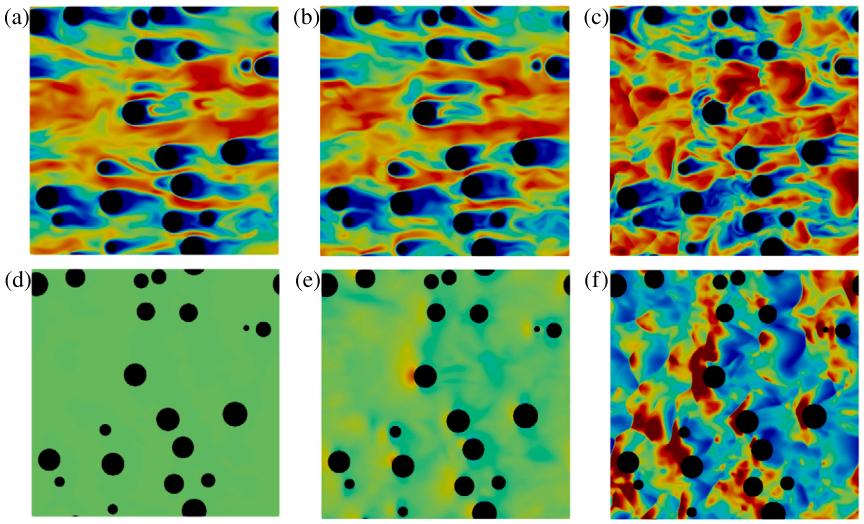


Fig. 7. PR-DNS of high-speed gas flow through a curtain of fixed particles with $\alpha_p = 0.1$ (Khalloufi and Capecelatro, 2023). Top row: magnitude of gas velocity. Bottom row: fluid pressure. Left column: $Ma = 0.1$. Middle column: $Ma = 0.5$. Right column: $Ma = 1$. Note that increasing Ma increases the mean gas velocity $\mathbf{u}_f$ and, hence, the slip velocity $\mathbf{u}_{pf}$. For supersonic flow, the large pressure fluctuations enhance the fluid drag on the particles, an effect that must be included at the mesoscale by making the drag correlation depend on Ma. Image courtesy of Mehdi Khalloufi and Jesse Capecelatro.

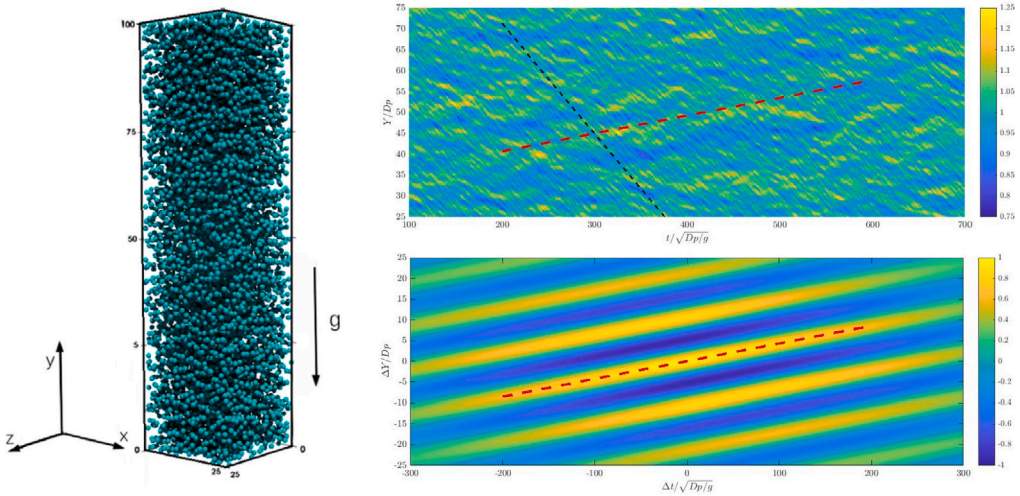


Fig. 8. PR-DNS of gravity-driven liquid–particle flow with density ratio $\mathcal{Z} = 2/3$, $\alpha_p = 0.3$, and $Ga = 144$ (Shajahan and Breugem, 2020, 2023). As particles fall with a constant average slip velocity (left), kinematic waves are produced such that $\langle \alpha_p \rangle$ depends on (t, y) where the average is taken over planes of constant y (top right). Using repeated auto-correlations, the speed and frequency of the kinematic waves (bottom right) can be determined from the PR-DNS data. The same analysis can be performed at the mesoscale to validate the model used for the pfp stresses. Images courtesy of Tariq Shajahan and Wim-Paul Breugem.

predicted kinematic wave *frequency* in the mesoscale and microscale simulations.

3.2. Mesoscale models

In the microscale simulations, the computational cost is mainly controlled by the need to resolve the fluid–particle interface in order to be able to accurately compute the stresses on the particle surface. For solid particles, a computational mesh with spacing of around $d_p/30$ is required where d_p is the particle diameter (Tenneti and Subramaniam, 2014). For bubbles and droplets, the grid requirements are even higher, even when automatic grid refinement is used at the interface (Farsoiya et al., 2023). Thus, the main motivation for developing a mesoscale model is to reduce the computational cost by modeling fluid–particle forces instead of resolving them. In practice, for mesoscale simulations the computational mesh spacing is on the order of d_p (Capecelatro and Desjardins, 2013), leading to a potential speed up of 30^3 for the same domain. However, the usual procedure is to increase the computational domain while keeping the total number of grid points constant. In this manner, with mesoscale models it is possible to simulate larger systems with the same computational cost as a microscale simulation.

3.2.1. Two types of mesoscale models

Broadly speaking, there are two categories of mesoscale models, one treats each particle with a Lagrangian approach and the other with an Eulerian approach based on a kinetic equation. In either case, the continuous fluid phase is treated with an Eulerian approach. Hereinafter, the former is referred to as the Euler–Lagrange approach, and the latter as the kinetic-based approach. In the literature, the Lagrangian approach is also called a discrete-element method (DEM) where the elements are the individual particles (Guo and Curtis, 2015). Likewise, in the literature the fluid-phase model may be solved on a coarse grid using a turbulence model for the unresolved scale. Such simulations are often referred to as CFD-DEM simulations (Nigmetova et al., 2022; Dufresne et al., 2020). In this work, only Euler–Lagrange models that fully resolve the fluid-phase equation (Capecelatro and

Desjardins, 2013) are considered. For example, for very dilute flows the fluid model reduces to the Navier–Stokes (NS) equation and is handled by a direct-numerical simulation. For denser cases, it will be necessary to model pseudoturbulence at the mesoscale because most of the fluid-phase fluctuations at the particle scale seen in PR-DNS will not be present at the mesoscale. This is a result of the fact that the fluid–particle interactions are treated with a drag coefficient based on the slip velocity.

With large computers, the Euler–Lagrange approach can be used with as many as 10^9 particles. For each particle, Newton's equations for linear and angular momentum are solved. This implies that the particle location and velocity are known for every particle. For dense flows, the main cost of the simulations is resolving the particle–particle interactions (e.g., collisions). While this is also true when solving the microscale model with PR-DNS, the results will not be identical because the fluid phase is not completely resolved at the mesoscale. For example, the pfp stresses are computed exactly at the microscale, but must be modeled at the mesoscale. Likewise, at the mesoscale the momentum-exchange terms such as drag, lift and added-mass forces must be modeled for each particle. However, because the particle locations and velocities are all known, the Euler–Lagrange models for these forces can be much more sophisticated (i.e., they can use two-particle or even multi-particle information Siddani and Balachandar, 2023) than the models used at the macroscale.

The kinetic model contains less information than the Euler–Lagrange model for the same system. This is because in the kinetic equation information concerning the locations of the individual particles is lost. This implies that the particle–particle collisions cannot be computed by following the particle trajectories. Instead, a kinetic model for collisions must be used that is based on knowledge of the one-particle velocity distribution function (VDF). The same observation applies to the mesoscale model for the pfp stresses for which a mean-field model is used. In the kinetic equation, the pfp stresses are modeled using information such as the slip velocity $\mathbf{u}_{pf}$ and particle volume fraction α_p. In principle, a larger time step can be employed to compute particle–particle collisions with a kinetic model, which lowers the computational cost compared to Euler–Lagrange simulations. Indeed, some 'coarse-grained' Euler–Lagrange models replace the exact collisions with a collision model to lower the computational cost.

In summary, the kinetic model contains less information than an Euler–Lagrange approach with resolved collisions. Nevertheless, the two mesoscale approaches have considerable overlap and can often be applied to the same polydisperse systems. As discussed in Section 3.3, one advantage of the kinetic approach is that it offers a pathway towards the macroscale model that is relatively straightforward to implement. Nonetheless, the ultimate success of the macroscale model depends strongly on the mesoscale closures developed for the microscale physics.

3.2.2. *Examples of the Euler–Lagrange approach*

In this section, two examples of the Euler–Lagrange approach applied to monodisperse multiphase flows are considered. The first example is a pressure-driven, gas–particle flow with gravity in a periodic domain (Capecelatro et al., 2015b). Compared to the PR-DNS results in Fig. 6 where the domain size was limited by the resolution requirement for resolving the boundary layers around the particles, at the mesoscale much larger domains are possible. As observed in Fig. 9, the falling particles form clusters that fall much faster than the slip velocity for a single particle. These clusters drag down the fluid inside them with nearly the same velocity,[2] while the fluid outside of the clusters moves upward to keep a domain-average fluid velocity of zero. Intuition might

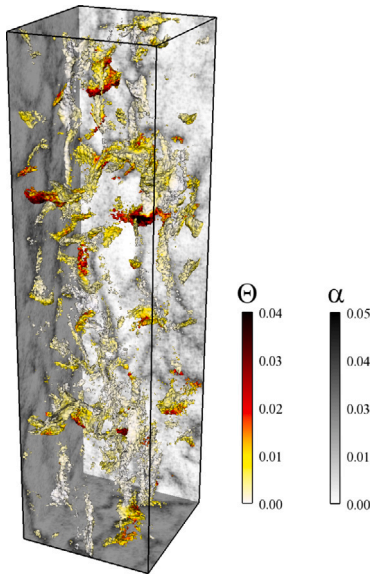


Fig. 9. Euler–Lagrange simulation of gravity-driven gas–particle flow in a periodic domain (Capecelatro et al., 2015b). The kinematic waves observed in the microscale model for liquid–solid flow now generate large 3-D clusters with a characteristic length many times larger than the particle diameter d_p. Compared to microscale simulations where a grid size of $d_p/30$ is required, at the mesoscale the grid size is on the order of the particle diameter. For the same computational cost, the domain used at the mesoscale can be 30 times larger than at the microscale, which allows enough space for the formation of clusters. Note that the granular temperature Θ is strongly correlated with the clusters. Image courtesy of Jesse Capecelatro.

suggest that the characteristic size of the clusters would continue to grow until all particles are in a single cluster. Indeed, this would minimize the overall drag. However, as the clusters grow, they form turbulent wakes that eventually interact with downstream clusters to break them up. For this reason, a statistical steady state is reached with constant slip velocity and mean cluster size.

It has been observed that the statistical properties depend on the computational domain size, but a plateau is eventually reached as the domain size increases (Capecelatro et al., 2015a). Increasing the domain size to reach the plateau is possible for mesoscale simulations, but not at the microscale. For this reason, it is very difficult to observe clusters in microscale simulations of gas–particles flows. The presence of clusters in experimental systems has been widely documented in the literature. In conclusion, it is interesting to note that the production of clusters is strongly correlated with the mass loading φ. For vertical turbulent channel flow (Capecelatro et al., 2016), clusters are observed for $\varphi > 1$ and the flow becomes laminar as φ increases from zero, before transitioning to cluster-induced turbulence for large φ (Capecelatro et al., 2018). Because $\rho_p \gg \rho_f$ in gas–particle flows, α_p does not have to be very large to reach $\varphi = 1$. For this reason, clusters are ubiquitous in pressure-driven gas–particle flows.

For the second example of Euler–Lagrange simulations, the liquid–particle flow in the microscale simulation in Fig. 8 is considered by repeating the same simulation using the mesoscale model in Capecelatro and Desjardins (2013) augmented with a model for the pfp stresses. As shown in Fig. 10 (as well as in the microscale results), clusters are observed in sedimentation problems even when the density ratio is not very small (i.e., $\mathcal{Z} = 2/3$). Compared to gas–particle flows, the clusters are smaller and can be resolved by microscale PR-DNS (Shajahan and Breugem, 2023). Thus, the main point of interest is to compare the mesoscale predictions directly with the microscale statistics. In particular, the magnitudes of the vertical and horizontal particle-velocity fluctuations as a function of the volume fraction $\phi = \alpha_p$ in Fig. 11 are of interest. It is well understood that the vertical component is larger than the horizontal component because the former has a source term due

[2] This 'cluster' added mass is reminiscent of the added mass of individual particles described earlier. Thus, in principle, it should included in a turbulence model for CIT.

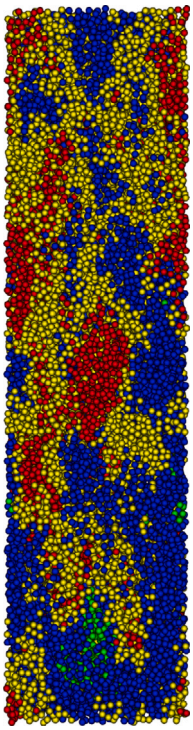


Fig. 10. Euler–Lagrange simulation of gravity-driven liquid–particle flow using the same parameters as for the PR-DNS in Fig. 8. The colors correspond to the particle velocity (see Shajahan and Breugem (2023)) and reveal the presence of clusters wherein all particles have nearly the same velocity. Compared to the gas–solid flow in Fig. 9, the cluster size is smaller, making it possible to observe clusters with PR-DNS. Image courtesy of Cheng Peng. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

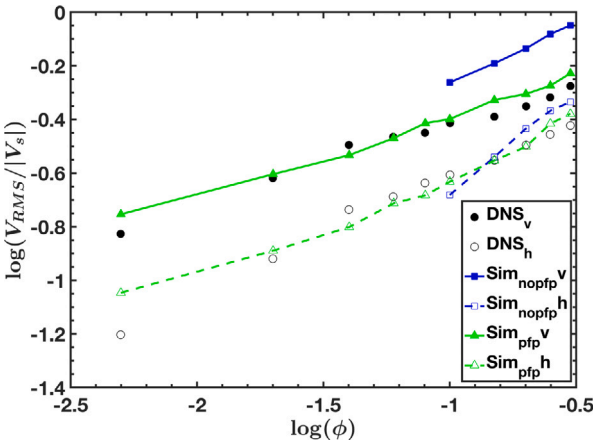


Fig. 11. Velocity fluctuations scaled by the slip velocity in the mesoscale Euler–Lagrange simulation of the gravity-driven liquid–particle flow in Fig. 8 (PR-DNS: symbols without lines). The vertical fluctuations are shown with filled symbols and the horizontal with empty symbols. A strong anisotropy in the vertical direction is seen for all ϕ. The blue curves correspond to the mesoscale results without the pfp stresses, and the green curves are with the pfp stresses. Both the magnitude of the fluctuations and their anisotropy are correctly reproduced when a mesoscale model for the pfp stresses is employed. Plot courtesy of Cheng Peng. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to fluid drag (Capecelatro et al., 2016). Thus, the velocity fluctuations in gravity-driven systems are always anisotropic. This can be observed from the PR-DNS results plotted in Fig. 11. Note that fluctuations in the two horizontal directions are equal, but lower than in the vertical direction. The particle-phase kinetic energy is found by summing the three components.

As noted earlier, the microscale PR-DNS resolves the pfp stresses while they must be modeled in the mesoscale simulation. If no model is used for the pfp stresses, the Euler–Lagrange simulations do not predict the correct magnitude and anisotropy for the velocity fluctuations. This is shown with the blue curves in Fig. 11 where it is clearly observed that compared to PR-DNS the particle-phase kinetic energy is too large. Note that most of this excess energy can be attributed to the velocity of the clusters (as opposed to the granular temperature of individual particles). In contrast, if a model for the pfp stresses is included in the Euler–Lagrange model, then (green curves) it is possible to match rather closely the PR-DNS results. For these results, the pfp-stress model has the same form as used in the kinetic model discussed in Section 3.2.3. Thus, it does not use any information concerning the individual particle positions and velocities that is available in the Euler–Lagrange approach. Physically, the pfp stresses damp out the particle velocity fluctuations due to long-range interactions. As seen below, in an Eulerian formulation, the pfp stresses enhance the particle-phase pressure to produce this effect.

Interestingly, the model for the pfp stresses also changes the wavelength of the kinematic waves observed in the Euler–Lagrange simulations. This effect is shown in Fig. 12. Without the pfp-stress model, the wavelength is shorter causing more rapid vertical oscillations in $\langle\alpha_p\rangle$ than seen with the pfp-stress model. This can again be attributed to the damping effect of the pfp pressure, which is larger when α_p is smaller, and vice versa. On the other hand, the velocity of the kinematic waves does not depend on the pfp stresses. Instead, it depends on the α_p dependence of the drag coefficient (which is well approximated by the Richardson–Zaki correlation Richardson and Zaki, 1954) used to determine the drag coefficient in the mesoscale model (Shajahan and Breugem, 2023).

In summary, as seen from the two example applications of the Euler–Lagrange approach, it is possible to simulate complex physics such as cluster-induced turbulence and kinematic waves that compare well with PR-DNS and/or experimental observations. However, in order to achieve good agreement for the particle-phase velocity fluctuations for the liquid–solid flows, it is necessary to include a model for the pfp stresses. This was expected because the pfp stresses are proportional to the liquid density ρ_f and thus will be even more important for bubbly flows. Whenever possible, the correlations used in the mesoscale model for drag, lift, pfp stresses, etc., should be derived from simulation data found with the microscale model. As shown next, the same correlations can often be utilized in the kinetic model.

3.2.3. *Kinetic models for polydisperse multiphase flow*

Unlike in the Euler–Lagrange approach, a kinetic model does not provide information concerning *individual* particles. Instead, it is a balance equation for the *average number* of particles with specific properties, e.g., velocity $\mathbf{u}$ and mass ξ, at a specific time and location. In the polydisperse cases considered here (Fox et al., 2024), this number density function (NDF), denoted by $n(t,\mathbf{x},\mathbf{u},\xi)$, has independent variables of time t, space $\mathbf{x}$, velocity $\mathbf{u}$, and mass ξ. Thus, the number concentration of particles at time t and location $\mathbf{x}$ with properties $(\mathbf{u},\xi)$ is equal to $n(t,\mathbf{x},\mathbf{u},\xi)\,\mathrm{d}\mathbf{u}\,\mathrm{d}\xi$. More generally, ξ will be a vector of particle properties (mass, composition, temperature, etc.), while for monodisperse particles with fixed mass the NDF reduces to a velocity distribution function (VDF) denoted by $n(t,\mathbf{x},\mathbf{u})$. For example, the Boltzmann equation is a kinetic model for the VDF wherein particle–particle interactions are modeled as hard-sphere collisions with the particle diameter much smaller than the average distance between particle centers. In the Euler–Lagrange approach, the locations and velocities of the colliding particles are known so that collision events are simulated directly. In contrast, in the kinetic approach their exact locations and velocities are unknown so that a collision model is needed. Likewise, for particle interactions through the fluid phase (e.g., pfp stresses), the

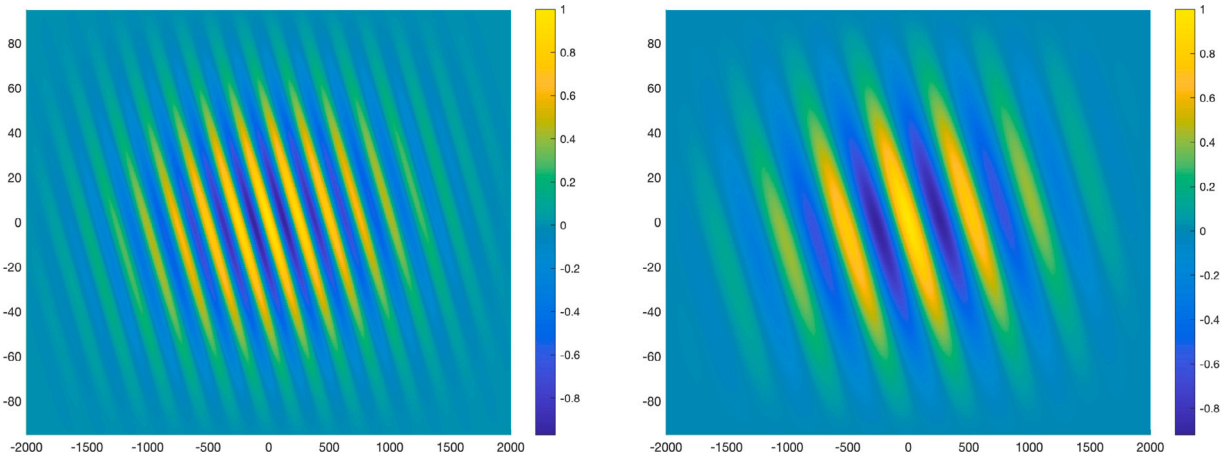


Fig. 12. Kinematic waves in the mesoscale Euler–Lagrange simulation of gravity-driven liquid–particle flow. Left: without pfp stresses. Right: with pfp stresses. The slopes of the correlation bars correspond to the kinematic wave velocity and do not depend on the pfp stresses, but the wavelength (distance between bars) changes with the magnitude of the pfp stresses. This dependence can be used to validate the model for the pfp stresses employed at the mesoscale. Images courtesy of Cheng Peng.

kinetic approach uses a model involving the spatial dependence of the NDF or its moments (e.g., the particle-phase volume fraction α_p and the slip velocity $\mathbf{u}_{fp}$).

Generally speaking (Marchisio and Fox, 2013), particle–particle collisions can end with a rebound (usually with loss of kinetic energy), or aggregation where the shapes of the two particles do not change when forming the new larger particle, or agglomeration where the particle shapes deform, or coalescence when the particles are a fluid. In any case, when particles stick together to form a larger size, the system will be polydisperse due to the resulting distribution of particle masses. The generic form of the kinetic equation governing the NDF is (Marchisio and Fox, 2013)

$$\frac{\partial n}{\partial t}+\mathbf{u}\cdot\frac{\partial n}{\partial \mathbf{x}}+\frac{\partial}{\partial \mathbf{u}}\cdot(\mathbf{A}n)+\frac{\partial}{\partial \xi}(Gn)=\text{Collisions}+\text{Aggregation}+\text{Breakage} \tag{1}$$

where aggregation can be replaced by whichever process occurs in the system of interest. The first term on the left-hand side is accumulation and the second is free transport with velocity $\mathbf{u}$. In this kinetic equation, the next term involving $\mathbf{A}$ represents acceleration due to body forces and fluid drag. The pfp stresses and a random source term due to PTKE may also be included in $\mathbf{A}$. The term involving G represents continuous growth of particle mass (e.g., due to surface reactions with the fluid phase). The terms on the right-hand side correspond to time-discontinuous processes. For example, if two particles collide they may bounce apart (inelastic collision) or stick together (aggregation), or break into multiple parts. Breakage can also occur due to fluid stresses on the particle surface.

As in the Euler–Lagrange approach, *mechanistic* models based on the microscale physics must be postulated for each of the unclosed terms in the kinetic equation. In fact, the physical models for the terms on the left-hand side of Eq. (1) will have the same form as in the Euler–Lagrange model. In any case, all the physical models in Eq. (1) must appear in closed form given the NDF or its moments, and must conserve the microscale physics (i.e., conservation of mass, momentum and energy for elastic collisions, etc.). For example, the model for hard-sphere collisions can invoke the *molecular-chaos assumption* to replace the two-particle NDF by the product of the VDF for the velocities of the two colliding particles. From the perspective of mesoscale modeling, the principal challenge is to formulate a kinetic equation that represents as closely as possible the microscale physics found by solving the microscale model. Compared to the Euler–Lagrange approach that resolves collisions, the main challenge at the kinetic level is to develop accurate closures for the terms on the right-hand side of Eq. (1). In practice, the accuracy of such closures can be validated by comparison with Euler–Lagrange results for the same system, as was done in Patel et al. (2017) and Baker et al. (2020) for turbulent gas–particle flow.

In order to account for added mass (Boniou et al., 2023), the definition of the NDF can be changed to include an added-mass factor β multiplied by the particle mass m_p, i.e., for monodisperse particles $f(t,\mathbf{x},\mathbf{u})=\beta(t,\mathbf{x})m_p n(t,\mathbf{x},\mathbf{u})$ where $\beta(t,\mathbf{x})=1+c_m\mathcal{Z}\alpha_f$ and $c_m(t,\mathbf{x})$ is the local added-mass constant. In the monodisperse case, the kinetic equation for f is

$$\frac{\partial f}{\partial t}+\mathbf{u}\cdot\frac{\partial f}{\partial \mathbf{x}}+\frac{\partial}{\partial \mathbf{u}}\cdot(\mathbf{A}f)=\text{Collisions}+\text{Added Mass} \tag{2}$$

where aggregation and breakage are null since they would yield polydisperse particles. The added-mass source term on the right-hand side represents fluid exchange between the particle wake and the continuous fluid phase (S_a in Fig. 4). As shown below, this term provides a source term in the transport equation for the added-mass factor β to reflect changes in the wake (i.e., to c_m) to adjust to different local fluid-phase volume fractions. In terms of the particle-mass concentration, the mass-weighted NDF is $f=m_p^\star n$ where $m_p^\star$ depends on $(t,\mathbf{x})$ through β.

For fluid–particle flows, the Navier–Stokes (NS) equation for the fluid phase is coupled with the kinetic equation for the particle phase to conserve overall mass, momentum and energy. In fact, only the moments of the NDF representing particle-phase mass, momentum and kinetic energy are required. The relevant moments for monodisperse particles are thus the following:

$$\rho_p\alpha_p=m_p\int n(\mathbf{u})\,\mathrm{d}\mathbf{u},\quad \rho_e\alpha_p^\star=\int f(\mathbf{u})\,\mathrm{d}\mathbf{u},\quad \rho_e\alpha_p^\star\mathbf{u}_p=\int f(\mathbf{u})\mathbf{u}\,\mathrm{d}\mathbf{u},$$
$$\rho_e\alpha_p^\star K_p=\int f(\mathbf{u})\frac{1}{2}u^2\,\mathrm{d}\mathbf{u}. \tag{3}$$

Given that the kinetic equations in Eqs. (1) and (2) are closed, they can in principle be solved to find $n(t,\mathbf{x},\mathbf{u})$ and $f(t,\mathbf{x},\mathbf{u})$. Thus, in principle, the integrals in Eq. (3) can be evaluated to find the particle-phase variables appearing in the fluid-phase balance equations.

Conceptually, this coupling procedure is equivalent to the volume-filtering procedure used in the Euler–Lagrange approach to estimate the particle-phase variables on the computational mesh (Capecelatro and Desjardins, 2013). In practice, for a given flow the 3-D computational mesh in $\mathbf{x}$-space needed for the fluid phase will be the same for the Euler–Lagrange and the kinetic models (e.g., on the order of the particle diameter). The direct solution of Eq. (2) to find $f(t,\mathbf{x},\mathbf{u})$ using a computational mesh in 6-D $(\mathbf{x},\mathbf{u})$-space is intractable for almost all applications. However, a Monte-Carlo approach such as the particle-in-cell (PIC) method (Birdsall and Langdon, 1985) can be used to approximate the NDF. For dilute systems where collisions are rare (i.e., the right-hand side of Eq. (2) is negligible), the PIC method is

equivalent to the Euler–Lagrange method, the only difference being that PIC uses notional particles with weights while the Euler–Lagrange method uses real particles (Capecelatro and Desjardins, 2013). However, for denser systems with frequent particle–particle collisions, the PIC method uses a collision model based on the closure appearing in the kinetic equation (Bird, 1994). Thus, the computational cost of the PIC method will be less than the Euler–Lagrange method because the latter requires a smaller time step to resolve collisions. However, the PIC results can only be as accurate as the collision model used in the kinetic equation. In general, collision models for non-spherical particles that can provide the same statistics as the Euler–Lagrange method have yet to be developed. Fortunately, for polydisperse flows with a very large number of particles, semi-empirical closures have been developed at the level of the balance equations for the lower-order moments given in Eq. (3) (Andries et al., 2002). Due to the large number of particle–particle collisions, such closures are usually insensitive to the exact nature of the individual collision events.

3.3. Macroscale model found from the kinetic model

As mentioned in the previous section, the coupling between the kinetic model for the particle phase and the NS equation for the fluid phase requires knowledge of the moments of the NDF. Starting from Eq. (1) and the definition of the moments in Eq. (3), it is relatively straightforward to derive *unclosed* balance equations for the moments. For example, the particle-phase mass balance for the monodisperse case is found from the kinetic equation by evaluating the following integral:

$$m_p \int \left\{ \frac{\partial n}{\partial t} + \mathbf{u} \cdot \frac{\partial n}{\partial \mathbf{x}} + \frac{\partial}{\partial \mathbf{u}} \cdot (\mathbf{A} n) = \text{Collisions} \right\} \mathrm{d}\mathbf{u}. \tag{4}$$

Using integration by parts, the integral reduces to

$$\frac{\partial \rho_p \alpha_p}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot (\rho_p \alpha_p \mathbf{u}_p) = m_p \int \{\text{Collisions}\} \, \mathrm{d}\mathbf{u} = 0 \tag{5}$$

where the right-hand side is zero due to conservation of mass during a collision.

Likewise, starting from Eq. (2), the following equation is found:

$$\frac{\partial \rho_e \alpha_p^\star}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot (\rho_e \alpha_p^\star \mathbf{u}_p) = \int \{\text{Added mass}\} \, \mathrm{d}\mathbf{u} = S_a \tag{6}$$

where S_a is the source term for added mass. Using the relation $\rho_e \alpha_p^* = \beta \rho_p \alpha_p$, the Eulerian balance for β can be found from Eqs. (5) and (6):

$$\frac{\partial \beta}{\partial t} + \mathbf{u}_p \cdot \frac{\partial \beta}{\partial \mathbf{x}} = \frac{S_a}{\rho_p \alpha_p} = \frac{1}{\tau_a} (\beta^\star - \beta) \tag{7}$$

where τ_a is the wake-relaxation timescale towards the equilibrium value $\beta^\star = 1 + c_m^\star \mathcal{Z} \alpha_f$. Here $c_m^\star(\alpha_p)$ is the equilibrium added-mass constant (e.g., for spherical particles $c_m^\star = 0.5$ as $\alpha_p \to 0$). Notice that in a PIC simulation of the kinetic model, β will be the same for all particles at a fixed location $\mathbf{x}$ and time t. Moreover, if τ_a is much smaller than the other characteristic timescales in the flow, to a good approximation $\beta^\star$ can be used in place of β to define the added mass. Nevertheless, care must be taken in the numerical algorithm to ensure mass conservation between the two phases when β changes the particle-phase added mass (Boniou and Fox, 2023).

For the particle-phase momentum balance, the derivation starts with Eq. (2) using the following integral:

$$\int \mathbf{u} \left\{ \frac{\partial f}{\partial t} + \mathbf{u} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{\partial}{\partial \mathbf{u}} \cdot (\mathbf{A} f) = \text{Collisions} + \text{Added Mass} \right\} \mathrm{d}\mathbf{u}. \tag{8}$$

Because collisions conserve momentum while generating a spatial flux, this equation reduces to

$$\frac{\partial}{\partial t}(\rho_e \alpha_p^\star \mathbf{u}_p) + \frac{\partial}{\partial \mathbf{x}} \cdot \left(\int \mathbf{u} \otimes \mathbf{u} f \, \mathrm{d}\mathbf{u} + \Pi_p \right) - \int \mathbf{A} f \, \mathrm{d}\mathbf{u} = \int \mathbf{u} \{\text{Added Mass}\} \, \mathrm{d}\mathbf{u} = \mathbf{S}_a \tag{9}$$

where Π_p is a second-order tensor representing the collisional flux and $\mathbf{S}_a$ is the added-mass momentum source term. The kinetic theory of granular flows (KTGF) can be used to find a closure for Π_p, which scales like α_p^2 (Gidaspow, 1994; Johnson and Jackson, 1987). For example, the trace of Π_p yields a collisional contribution to the granular pressure:

$$P_c = 2 \rho_e \alpha_p^\star (1 + e_c) \alpha_p g_0(\alpha_p) \Theta_p$$

where e_c is the coefficient of restitution and g_0 the two-particle correlation function. The added-mass momentum source term depends on the sign of S_a (i.e., mass transfer to/from the wake): $\mathbf{S}_a = \max(S_a, 0)\mathbf{u}_f + \min(S_a, 0)\mathbf{u}_p$. The integral involving the second moment of velocity can be expanded as

$$\int \mathbf{u} \otimes \mathbf{u} f \, \mathrm{d}\mathbf{u} = \rho_e \alpha_p^\star (\mathbf{u}_p \otimes \mathbf{u}_p + \boldsymbol{\sigma}_p)$$

where $\boldsymbol{\sigma}_p$ is the covariance tensor for the particle velocity. The trace of $\boldsymbol{\sigma}_p$ is equal to $3\Theta_p$ and, hence, the kinetic contribution to the granular pressure is $P_k = \rho_e \alpha_p^\star \Theta_p$. Finally, the integral involving the acceleration term

$$\int \mathbf{A} f \, \mathrm{d}\mathbf{u}$$

will usually not be in closed form due to the nonlinear dependence of the drag coefficient on $\mathbf{u}$. In the context of solving the kinetic equation using a PIC method (Birdsall and Langdon, 1985), a statistical approximation of the NDF will be available at each computational grid cell. Using this estimate, the unclosed integral terms in Eq. (9) can be computed at each time step.

In summary, for a given kinetic model for the particle phase, it is always possible to find unclosed balance equations for the quantities needed for coupling with the NS equation for the fluid phase. Because these moment equations inherit the properties of the kinetic model, they will conserve mass, momentum and energy. In order to control statistical fluctuations present in Monte-Carlo methods, solving the moment balance equations for the particle-phase mass, momentum and energy can be useful instead of employing the estimates provided by the PIC-estimate of the NDF (Birdsall and Langdon, 1985). When applicable, a powerful alternative to Monte-Carlo methods is to close the moment equations to arrive at a macroscale model. In general, solving the macroscale model numerically will be computationally less expensive than solving the mesoscale model. This is especially true when the physics of the particle phase are dominated by particle–particle collisions, in which case the KTGF provides good closures at the level of the NS equation (i.e., mass, momentum, and energy). In contrast, for very dilute flows the closure of the velocity moments (e.g., $\boldsymbol{\sigma}_p$) is highly flow dependent. Thus, as done for rarefied gas flows, direct-simulation Monte-Carlo (DSMC) methods (Bird, 1994) may be needed to solve the mesoscale kinetic model for such flows.

4. Mesoscale models for gas–particle flow

As shown in the previous section, a mesoscale model for gas–particle flow can be written as either an Euler–Lagrange model or as a kinetic model. In this section, specific examples of kinetic models are presented for gas–particle flow with added mass, starting with the monodisperse case. The kinetic model governs the mass-weighted NDF $f(t, \mathbf{x}, \mathbf{u})$ including added mass (Boniou et al., 2023), which is defined by

$$f(t, \mathbf{x}, \mathbf{u}) = \beta(t, \mathbf{x}) m_p n(t, \mathbf{x}, \mathbf{u}) = m_p^\star n(t, \mathbf{x}, \mathbf{u}).$$

Here, as before, $n(t, \mathbf{x}, \mathbf{u})$ is the number-based NDF, the added-mass factor $\beta(t, \mathbf{x})$ is governed by Eq. (7), and m_p is the mass of a particle. Thus, $m_p^\star$ is the mass of a particle plus its added mass, which depends on $(t, \mathbf{x})$ through β.

4.1. Mesoscale model for monodisperse particles

An example kinetic model for the NDF $f(t,\mathbf{x},\mathbf{u})$ is (Boniou et al., 2023)

$$\frac{\partial f}{\partial t}+\frac{\partial}{\partial \mathbf{x}}\cdot\left(\mathbf{u}f-P_p\frac{\partial f}{\partial \mathbf{u}}\right)+\frac{\partial}{\partial \mathbf{u}}\cdot\left(\mathbf{A}f-\frac{f}{\rho_e}\frac{\partial \hat{p}_f}{\partial \mathbf{x}}-\frac{f}{\rho_e\alpha_p^\star}\frac{\partial}{\partial \mathbf{x}}\cdot\mathbf{P}_{pfp}\right)$$
$$=\frac{\partial^2}{\partial \mathbf{u}\partial \mathbf{u}}:(\mathbf{B}_u f)+C+F+S. \tag{10}$$

At the mesoscale, by construction, this equation is closed (i.e., it can be solved given initial and boundary conditions to find the NDF). The different terms appearing in the kinetic model and their physical meanings are as follows:

- The accumulation $\partial_t f$ and free-transport term $\partial_{\mathbf{x}}\cdot(\mathbf{u}f)$ represent the convection of the NDF at the phase-space velocity $\mathbf{u}$. In a Lagrangian framework, they correspond to the change of particle location $\mathrm{d}\mathbf{x}_o/\,\mathrm{d}t=\mathbf{u}_o$ where $\mathbf{u}_o$ is the velocity of a particle whose center of mass is located at $\mathbf{x}_o$.
- The scalar $P_p=P_c+P_{fr}$ (units $\mathrm{m}^2/\mathrm{s}^2$) is the sum of the normalized collisional and frictional (or close-packed) pressures.[3] Thus, the term involving P_p in Eq. (10) is a model for the spatial flux of f due to interactions between finite-size particles with different velocities. The KTGF models (Boniou and Fox, 2023) for these pressures are

  $$P_c=2(1+e_c)\alpha_p g_0(\alpha_f)\Theta_p \quad\text{and}\quad P_{fr}=\frac{p_{fr}\alpha_p g_0(\alpha_f)}{\rho_e\alpha_p^\star}h_{fr}(\alpha_p).$$

  The smoothed step function

  $$h_{fr}=\frac{1}{2}\left[1+\tanh\left(\frac{\alpha_p-\alpha_{max}}{\Delta_{fr}}\right)\right]$$

  determines the range of α_p values over which particles are in sustained contact. Thus, for $\alpha_p<\alpha_{max}$, P_{fr} is negligible. Physically, P_{fr} is the equation of state for the particle phase with sustained contacts and constant ρ_p. The model constant p_{fr} (units Pa) controls the resistance of particles to deformation (at fixed ρ_p) as α_p increases above α_{max}. The two-particle correlation function is modeled by the Carnahan–Starling approximation (Carnahan and Starling, 1969):

  $$g_0=\frac{1+\alpha_f}{2\alpha_f^3}$$

  and represents the increasing probability of collisions with decreasing α_f. An equation for $P_{fr}(\alpha_p)$ that increases monotonically with α_p will also be needed for monodisperse bubbly flows (i.e., a foam of bubbles that do not coalesce Stewart and Hilgenfeldt, 2023) or dense suspensions (Morris, 2020). In addition to the normal stresses, viscous stresses will be important in low-speed flows. For clarity, the viscous terms have not been included in the kinetic model, but are available from the KTGF (Gidaspow, 1994; Jenkins and Savage, 1983; Chauchat, 2018). Such terms are negligible in high-speed flows, but will be important for low-speed flows such as in gas–solid risers, fluidized beds, and dense suspensions.
- The acceleration model in Eq. (10):

  $$\mathbf{A}=\frac{1}{\tau_p}(\mathbf{u}-\mathbf{u}_f)+\mathbf{g}$$

  represents the contributions from the fluid drag and body force (gravity). Other forces, such as lift, can be added on the right-hand side when required by the flow physics. The drag timescale τ_p depends on the particle Reynolds number, defined by

  $$Re_p=\frac{\alpha_f^\star}{\alpha_f}\frac{\rho_f d_p u_{pf}}{\mu_f}$$

  for particle diameter d_p, and on α_p through the drag coefficient C_D (Tavanashad et al., 2021):

  $$\tau_p=\frac{4\rho_e d_p^2}{3\mu_f C_D Re_p}.$$

  In high-speed flow, C_D also depends on the particle Mach number (Khalloufi and Capecelatro, 2023; Osnes et al., 2023). Note that the slip velocity $\mathbf{v}_{pf}=\mathbf{u}_p-\mathbf{v}_f$ based on the average fluid-phase velocity found from $\alpha_f\mathbf{v}_f=\alpha_f^\star\mathbf{u}_f+\alpha_a\mathbf{u}_p$ is $\mathbf{v}_{pf}=\alpha_f^\star\mathbf{u}_{pf}/\alpha_f$ (Fox et al., 2020). For this reason, the ratio $\alpha_f^\star/\alpha_f$ appears in Re_p to be consistent with existing drag correlations that use a Reynolds number based on $\mathbf{v}_{pf}$.
- The next term in Eq. (10) involving the gas-phase pressure gradient is due to the Archimedes' (buoyancy) force. The PTKE pressure contribution is included in the definition $\hat{p}_f=p_f+\frac{2}{3}\rho_f k_f$ where p_f is the gas pressure found from an equation of state.
- The term in Eq. (10) involving the symmetric, second-order tensor $\mathbf{P}_{pfp}$ is due to the pfp stresses, which accelerate the particles. The mesoscale model for the pfp-pressure tensor is

  $$\mathbf{P}_{pfp}=C_{pfp}\rho_f\alpha_p^\star(C_1u_{pf}^2\mathbf{I}+C_2\mathbf{u}_{pf}\otimes\mathbf{u}_{pf}),$$

  and depends on the slip velocity $\mathbf{u}_{pf}=\mathbf{u}_p-\mathbf{u}_f$ (Wang et al., 2021; Bulthuis et al., 1995). The parameters C_1 and C_2, which depend on Re_p and α_p, control the relative contributions of the isotropic and anisotropic parts of $\mathbf{P}_{pfp}$ (i.e., $3C_1+C_2=1$), while $C_{pfp}>0$ controls the magnitude of its trace. Note that $\mathbf{P}_{pfp}$ scales with $\rho_f u_{pf}^2$ and, thus, can be significant in high-speed gas–particle flows even when $\rho_f\ll\rho_p$.
- The diffusion term in velocity space in Eq. (10) involving $\mathbf{B}_u$ arises from the random acceleration of particles due to fluid-phase PTKE (i.e., random forcing without mass exchange) and particle-velocity fluctuations (Boniou et al., 2023). The diffusion tensor is modeled as

  $$\mathbf{B}_u=\frac{1-a}{\tau_p}\left((\mathbf{u}-\mathbf{u}_p)\otimes(\mathbf{u}-\mathbf{u}_p)+\frac{2}{3}k_f\mathbf{I}\right)$$
  $$\text{with}\quad trace(\mathbf{B}_u)=\frac{1-a}{\tau_p}(u^2-2\mathbf{u}_p\cdot\mathbf{u}+u_p^2+2k_f)$$

  where k_f is the PTKE and $0.5<a<1$ is a model constant depending on $\mathcal{Z}$ (Tavanashad et al., 2019). For gas–particle flow, $a\approx 1$ due to the large inertia of the particles relative to the gas. Note that the diffusion in velocity space affects the kinetic energy but not the momentum of the particle phase. The expression for $\mathbf{B}_u$ must therefore be consistent with the drag-exchange model used for the fluid-phase PTKE. For the particle-phase kinetic energy balance, only the trace of $\mathbf{B}_u$ is needed.
- The particle–particle collision term in Eq. (10) is modeled using the BGK-like expression (Bhatnagar et al., 1954):

  $$C=\frac{1}{\tau_c}(f_G-f)$$

  where the collision timescale is

  $$\tau_c=\frac{d_p\sqrt{\pi}}{12\alpha_p g_0\Theta_p^{1/2}}$$

  and f_G is a Gaussian distribution with the same moments up to second order as f. For inelastic collisions, the moments of f_G with order higher than one depend on the coefficient of restitution e_c due to particle-phase kinetic energy decay (Passalacqua et al., 2011; Marchisio and Fox, 2013).

[3] The frictional pressure resists compaction of close-packed particles with constant material density. This resistance could be frictional, but can also be due to deformation of the particle shape (e.g., bubbles).

- The particle–particle friction term in Eq. (10) is modeled using the BGK-like expression (Boniou et al., 2023):

$$F = \frac{1}{\tau_{fr}}[\rho_e \alpha_p^\star \delta(\mathbf{u}-\mathbf{u}_p) - f]$$

where $\tau_{fr} = c_{fr}\tau_c/h_{fr}$ is the frictional timescale. Physically, friction drives the granular temperature Θ_p to zero while conserving mass and momentum. As done with P_{fr}, the timescale τ_{fr} depends on h_{fr} such that F is negligible unless α_p is above the close-packed limit α_{max}.

- The added-mass term S in Eq. (10), arising due to the exchange of mass, momentum and kinetic energy between the particle wakes and the continuous gas phase, is defined as (Boniou et al., 2023)

$$S = C_\xi f + C_u \frac{\partial}{\partial \mathbf{u}} \cdot [(\mathbf{u}-\mathbf{u}_f) f] + \frac{1}{2} C_u \frac{\partial^2}{\partial \mathbf{u}\partial \mathbf{u}} : (\mathbf{D}_u f)$$

with rate parameters (units 1/s)

$$C_\xi = \frac{S_a}{\rho_e \alpha_p^\star} = \frac{1}{\beta \tau_a}(\beta^\star - \beta) \quad \text{and} \quad C_u = \frac{1}{2}(C_\xi + |C_\xi|).$$

The velocity diffusion tensor is modeled as

$$\mathbf{D}_u = (\mathbf{u}-\mathbf{u}_f)\otimes(\mathbf{u}-\mathbf{u}_f) + \frac{2}{3}k_f \mathbf{I} \quad \text{with}$$
$$trace(\mathbf{D}_u) = u^2 - 2\mathbf{u}_f\cdot\mathbf{u} + u_f^2 + 2k_f.$$

Notice that C_u is null when $C_\xi \le 0$, which corresponds physically to the added mass from the particle wake moving to the continuous fluid phase. The second term on the right-hand side of S represents the change in the particle-phase momentum due to the addition of gas with velocity $\mathbf{u}_f$ to the wake, and the diffusion term represents the corresponding change in kinetic energy. For the particle-phase kinetic energy balance, only the trace of $\mathbf{D}_u$ is needed.

For clarity, in Eq. (10) the change of internal energy of the particles due, for example, to added mass, inelastic collisions and friction have been ignored (See Boniou et al. (2023) for the full model.). Thus, it must be assumed that the particle-phase kinetic energy lost during these processes is transferred directly to the gas phase. If the change of internal energy of the particles is important, then it can be included as an independent variable in the NDF and in the kinetic equation. Indeed, in the kinetic-based approach for modeling disperse multiphase flows, any new physical process must be included in the kinetic model. This is done through a mechanistic model for the particle-scale physics that has been validated using microscale simulation data. Now that all of the terms in Eq. (10) have been defined in closed form, the next step is to derive the moment balance equations for mass, momentum, and kinetic energy of the particle phase. This is done in Section 5.

4.2. Extension of mesoscale model to polydisperse particles

For clarity, the discussion of polydisperse particles is limited to the case where particles are spherical with different masses, denoted by ξ (Fox et al., 2024). Analogous to the monodisperse case in Section 4.1, the number concentration of particles with velocity $\mathbf{u}$ and mass ξ is defined as $n(t,\mathbf{x},\mathbf{u},\xi)\,d\mathbf{u}\,d\xi$ where the particle mass is related to the particle diameter d_p by $\xi = \rho_p k_v d_p^3$ with constant shape factor $k_v = \pi/6$. The zero-order moment $M_0(t,\mathbf{x}) = \int n(t,\mathbf{x},\mathbf{u},\xi)\,d\mathbf{u}\,d\xi$ is the particle number concentration including all sizes, while $M_1(t,\mathbf{x}) = \int \xi n(t,\mathbf{x},\mathbf{u},\xi)\,d\mathbf{u}\,d\xi$ is the particle mass concentration. The latter is related to the particle-phase volume fraction by $M_1 = \rho_p \alpha_p$. In addition, for the polydisperse case, the mass-weighted higher-order moments are defined by $M_k = \int \xi^k n(\mathbf{u},\xi)\,d\mathbf{u}\,d\xi$.

For the polydisperse kinetic model (Fox et al., 2024), the mass-weighted NDF with added mass is defined by[4]

$$f(t,\mathbf{x},\mathbf{u},\xi) = \beta(t,\mathbf{x})\,\xi\,n(t,\mathbf{x},\mathbf{u},\xi)$$

where the added-mass factor $\beta(t,\mathbf{x})$ is governed again by Eq. (7). As in the monodisperse case, the relationship between the different NDF is

$$\rho_e \alpha_p^\star = \int f(\mathbf{u},\xi)\,d\mathbf{u}\,d\xi = \beta \int \xi\, n(\mathbf{u},\xi)\,d\mathbf{u}\,d\xi = \beta\,\rho_p \alpha_p.$$

Thus, to define $\beta(t,\mathbf{x})$, it again suffices to solve the balance equations for $\rho_e\alpha_p^\star$ and $\rho_p\alpha_p$. The higher-order mass moments, denoted by

$$\mathcal{M}_{k+1} = \int \xi^k f(\mathbf{u},\xi)\,d\mathbf{u}\,d\xi = \beta \int \xi^{k+1} n(\mathbf{u},\xi)\,d\mathbf{u}\,d\xi,$$

are related to the moments M_k by $\mathcal{M}_k = \beta M_k$. Thus, working with either M_k or $\mathcal{M}_k$ provides equivalent information.

For a polydisperse system with particles that do not change size, the kinetic equation for $f(t,\mathbf{x},\mathbf{u},\xi)$ is (Fox et al., 2024)

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial \mathbf{x}}\cdot\left(\mathbf{u}f - P_p(\xi)\frac{\partial f}{\partial \mathbf{u}}\right) + \frac{\partial}{\partial \mathbf{u}}\cdot\left(\mathbf{A}(\xi)f - \frac{f}{\rho_e}\frac{\partial \hat{p}_f}{\partial \mathbf{x}} - \frac{f}{\rho_e\alpha_p^\star}\frac{\partial}{\partial \mathbf{x}}\cdot\mathbf{P}_{pfp}\right)$$
$$= \frac{\partial^2}{\partial \mathbf{u}\partial \mathbf{u}} : [\mathbf{B}_u(\xi) f] + C(\xi) + F(\xi) + S(\xi), \tag{11}$$

which has the same form as Eq. (10) for the monodisperse case. Thus, the physical significance of the terms in Eq. (11) is the same as in Eq. (10). However, the actual models can now depend on the particle mass ξ, as well as on the velocity $\mathbf{u}$ (Fox et al., 2024). For example, the drag timescale $\tau_p(\xi)$ has a strong dependence on the particle diameter d_p such that particles of different sizes can segregate due to fluid drag. For particles that change size due to growth, aggregation and/or breakage there will be additional closed terms in Eq. (11) that can be derived from the microscale model (Marchisio and Fox, 2013). For example, the continuous growth of a particle due to mass transfer from the continuous phase generates a new term on the left-hand side of Eq. (11) as seen in Eq. (1) where $G(\mathbf{u},\xi)$ is the growth rate of particles with velocity $\mathbf{u}$ and mass ξ. Likewise, aggregation and/or breakage will generate integral terms on the right-hand side of Eq. (11).

Regardless of the physics included in the kinetic model, it is important to recall that a well-posed kinetic model must be in closed form and thus Eq. (11) can be solved to find $f(t,\mathbf{x},\mathbf{u},\xi)$. However, the direct-numerical solution to find the NDF $f(t,\mathbf{x},\mathbf{u},\xi)$ is very challenging because of the number of independent variables (i.e., the eight variables $(t,\mathbf{x},\mathbf{u},\xi)$). As in the monodisperse case, a statistical method such as PIC (Birdsall and Langdon, 1985) or DSMC (Bird, 1994) can be employed to approximate the NDF using notional particles. In any case, the fluid-phase NS equation must be solved simultaneously and is coupled to the kinetic model through the moments of the NDF. Thus, when applicable, it is usually preferable to develop a macroscale model starting from the mesoscale model that uses as few moments as possible (but no less!) to capture the relevant flow physics.

5. Macroscale models for gas–particle flow

In Section 3.3, a brief overview was presented on how a macroscale model is found from a mesoscale kinetic equation. Here, through the specific example of gas–particle flow from Section 4, further details are provided on how closure of the velocity moment equations leads to a closed macroscopic model. In addition, the issue of well-posed models is discussed, as well as how this property is verified using the macroscopic model equations for the particle and fluid phases. In the context of kinetic-based modeling, it is important to recognize that a well-posed mesoscale model can lead to an ill-posed macroscale model.

[4] Alternatively, an NDF defined as $\beta(t,\mathbf{x})\,n(t,\mathbf{x},\mathbf{u},\xi)$ could be used (Fox et al., 2024). The only difference is that the order of the mass moments differs by one compared to $f(t,\mathbf{x},\mathbf{u},\xi)$.

This is because at the macroscale one does not have direct access to the NDF, only to some of its moments. In summary, obtaining a well-posed mesoscale model depends on the *physical models* used to describe the microscale physics, while a well-posed macroscale model depends on the *mathematical approximation* of the unknown NDF.

5.1. Macroscale model for monodisperse particles

For monodisperse gas–particle flow, the moment equations used in the macroscale model are found from the kinetic model in Eq. (10) by integrating over velocity phase space. Here, consider the following moments corresponding to mass, momentum and kinetic energy (Boniou et al., 2023):

- For the mass balances, two zero-order moments are required:

  $$\rho_p\alpha_p = m_p\int n\,\mathrm{d}\mathbf{u} \quad \text{and} \quad \rho_e\alpha_p^\star = \int f\,\mathrm{d}\mathbf{u}.$$

  The kinetic equation then yields

  $$\partial_t(\rho_p\alpha_p) + \partial_{\mathbf{x}}\cdot(\rho_p\alpha_p\mathbf{u}_p) = 0 \tag{12}$$

  and

  $$\partial_t(\rho_e\alpha_p^\star) + \partial_{\mathbf{x}}\cdot(\rho_e\alpha_p^\star\mathbf{u}_p) = S_a. \tag{13}$$

  These equations are closed.
  The corresponding mass balance for the gas phase is

  $$\partial_t(\rho_f\alpha_f^\star) + \partial_{\mathbf{x}}\cdot(\rho_f\alpha_f^\star\mathbf{u}_f) = -S_a. \tag{14}$$

  Notice that the sum of mass balances for each phase has no terms on the right-hand side so that the mixture density, defined by $\rho_m = \rho_e\alpha_p^\star + \rho_f\alpha_f^\star = \rho_p\alpha_p + \rho_f\alpha_f$, is conserved. Also note that Eq. (12) can be replaced by the transport equation for β in Eq. (7), which provides the same information.

- For the particle-phase momentum, the first-order moment $\rho_e\alpha_p^\star\mathbf{u}_p = \int \mathbf{u}f\,\mathrm{d}\mathbf{u}$ is required. For this moment, the kinetic equation yields

  $$\partial_t(\rho_e\alpha_p^\star\mathbf{u}_p) + \partial_{\mathbf{x}}\cdot[\rho_e\alpha_p^\star(\mathbf{u}_p\mathbf{u}_p + \boldsymbol{\sigma}_p + P_p\mathbf{I}) + \mathbf{P}_{pfp}] = \frac{\rho_e\alpha_p^\star}{\tau_p}(\mathbf{u}_f - \mathbf{u}_p) - \alpha_p^\star\partial_x\hat{p}_f + \mathbf{S}_a + \rho_e\alpha_p^\star\mathbf{g} \tag{15}$$

  where the added-mass source term for momentum is

  $$\mathbf{S}_a = \int \mathbf{u}\,S\,\mathrm{d}\mathbf{u} = \max(S_a,0)\,\mathbf{u}_f + \min(S_a,0)\,\mathbf{u}_p.$$

  The unclosed term in the particle-phase momentum balance equation is the velocity covariance $\boldsymbol{\sigma}_p$. For collisional high-speed flow, the isotropic closure $\boldsymbol{\sigma}_p = \Theta_p\mathbf{I}$ with granular temperature $3\Theta_p = 2K_p - u_p^2$ can be employed (Fox, 2014). For more dilute flows, it is possible to solve the balance equation for the second-order tensor $\boldsymbol{\sigma}_p$ (Vié et al., 2015). To complete the isotropic closure, a transport equation for the particle-phase kinetic energy K_p is needed.
  The corresponding momentum balance for the gas phase is

  $$\partial_t(\rho_f\alpha_f^\star\mathbf{u}_f) + \partial_{\mathbf{x}}\cdot(\rho_f\alpha_f^\star\mathbf{u}_f\mathbf{u}_f + \hat{p}_f\mathbf{I}) = \frac{\rho_e\alpha_p^\star}{\tau_p}(\mathbf{u}_p - \mathbf{u}_f) + \alpha_p^\star\partial_{\mathbf{x}}\hat{p}_f - \mathbf{S}_a + \rho_f\alpha_f^\star\mathbf{g}. \tag{16}$$

  Notice that the sum of the momentum balances for each phase has only gravity on the right-hand side so that the mixture momentum, defined by $\rho_m\mathbf{u}_m = \rho_e\alpha_p^\star\mathbf{u}_p + \rho_f\alpha_f^\star\mathbf{u}_f$, is conserved.

- For the particle-phase kinetic energy, the second-order moment $\rho_e\alpha_p^\star K_p = \int \frac{1}{2}u^2 f\,\mathrm{d}\mathbf{u}$ is required. For this moment, the kinetic equation yields

  $$\partial_t(\rho_e\alpha_p^\star K_p) + \partial_{\mathbf{x}}\cdot[\rho_e\alpha_p^\star(K_p + \Theta_p + P_p)\mathbf{u}_p + \mathbf{P}_{pfp}\cdot\mathbf{u}_p] = \mathbf{P}_{pfp} : (\partial_{\mathbf{x}}\mathbf{u}_p) + D_K - \alpha_p^\star\mathbf{u}_p\cdot\partial_{\mathbf{x}}\hat{p}_f + S_K + H_p + \rho_e\alpha_p^\star\mathbf{u}_p\cdot\mathbf{g} \tag{17}$$

  where the added-mass source term for kinetic energy is

  $$S_K = \int \frac{1}{2}u^2\,S\,\mathrm{d}\mathbf{u} = \max(S_a,0)\,K_f + \min(S_a,0)\,K_p.$$

  The first term on the right-hand side of Eq. (17) involving the pfp-pressure tensor represents the compression work done on the particle phase (i.e., it is positive when pfp stresses increase α_p). The fluid drag and collisional/frictional terms (i.e., $C + F$) yield energy-dissipation/exchange terms:

  $$D_K = \int \frac{1}{\tau_p}\left[\frac{1}{2}\left(\mathbf{u}\cdot\mathbf{u}_f - u^2\right) + (1-a)\,trace(\mathbf{B}_u)\right] f\,\mathrm{d}\mathbf{u} = -\frac{\rho_e\alpha_p^\star}{\tau_p}[3a\Theta_p - 2(1-a)k_f + \mathbf{u}_{pf}\cdot\mathbf{u}_p]$$

  and

  $$H_p = \int \frac{1}{2}u^2(C+F)\,\mathrm{d}\mathbf{u} = -\rho_e\alpha_p^\star\left[\frac{1}{\tau_p}(1-e_c^2) + \frac{1}{\tau_{fr}}\right]\Theta_p$$

  where H_p is negative. Notice that D_K has a component proportional to k_f, which represents an exchange of fluid-phase PTKE to generate particle-phase kinetic energy. Likewise, the term in D_K proportional to Θ_p represents the loss of particle-phase kinetic energy to produce PTKE in the fluid phase. The value of the energy-exchange constant a, which controls the equilibrium energy ratio $3\Theta_p/(2k_f) = (1-a)/a$, is found from

  $$\frac{1-a}{1-a_{min}} = \frac{\mathcal{Z}}{K+\mathcal{Z}}.$$

  The parameters $a_{min} = 0.5$ and $K = 0.06$ were fit for $Z = \infty$ and $Z = 1$, respectively, using the PR-DNS data from Tavanashad et al. (2019). This constant determines the dominant direction of the energy exchange between the two phases for a given density ratio (Fox et al., 2020). Finally, to be consistent with the momentum balance, the third-order velocity moment in the spatial flux has been closed using $\Theta_p\mathbf{u}_p$, and therefore it is assumed that the third-order central moments are null. This is equivalent to assuming that the VDF is Maxwellian. However, if particle–particle collisions are infrequent and/or the Stokes number is not small, the spatial flux of particle-phase momentum and kinetic energy should be closed using higher-order velocity moments (Fox and Laurent, 2022).
  The corresponding total-energy balance for the gas phase is

  $$\partial_t(\rho_f\alpha_f^\star E_f) + \partial_{\mathbf{x}}\cdot[\rho_f\alpha_f^\star E_f\mathbf{u}_f + (\alpha_f^\star\mathbf{u}_f + \alpha_p^\star\mathbf{u}_p)\hat{p}_f] = -\mathbf{P}_{pfp} : (\partial_{\mathbf{x}}\mathbf{u}_p) - D_K + \alpha_p^\star\mathbf{u}_p\cdot\partial_{\mathbf{x}}\hat{p}_f - S_K - H_p + \rho_f\alpha_f^\star\mathbf{u}_f\cdot\mathbf{g}. \tag{18}$$

  The exact spatial flux term on the left-hand side was derived from kinetic theory (Fox, 2019). It depends on the volume-average velocity $(\alpha_f^\star\mathbf{u}_f + \alpha_p^\star\mathbf{u}_p)$, and not just on $\mathbf{u}_f$. Notice that the sum of the energy balances for each phase has only gravity on the right-hand side so that the overall total energy, defined by $\rho_m E_m = \rho_e\alpha_p^\star E_p + \rho_f\alpha_f^\star E_f$, is conserved. (Recall that for simplicity, the particle-phase internal energy e_p is assumed to be constant, thereby neglecting the change due to added-mass exchange from the fluid.) Also, notice that in the gas phase H_p generates a positive source term that increases the fluid-phase internal energy.
  In order to complete the macroscale model, a model for fluid-phase PTKE is needed (Shallcross et al., 2020). A simple model that captures most of the important physics is

  $$\partial_t(\rho_f\alpha_f^\star k_f) + \partial_{\mathbf{x}}\cdot(\rho_f\alpha_f^\star k_f\mathbf{u}_f) = -\frac{2}{3}\rho_f\alpha_f^\star k_f\partial_{\mathbf{x}}\cdot\mathbf{u}_f + \frac{\rho_e\alpha_p^\star}{\tau_p}[3a\Theta_p - 2(1-a)k_f + u_{pf}^2 - C_f k_f] \tag{19}$$

  wherein the drag-production term appears inside the second term on the right-hand side. The first term on the right-hand side is

a production term in the fluid phase and has been closed by replacing the PTKE Reynolds-stress tensor with an isotropic tensor with the same PTKE. The final term involving the parameter C_f, defined by

$$C_f^{-1} = \alpha_p[1 + 1.25\alpha_f^3 \exp(-\alpha_p \alpha_f^{1/2} Re_p^{1/2})],$$

represents PTKE dissipation to produce fluid-phase internal energy e_f. Notice that the term on the right-hand side of Eq. (19) involving u_{pf}^2 results from assuming that all of the mean kinetic energy lost by the particle phase ends up as k_f (see Fig. 5), and is subsequently dissipated in the fluid phase to increase e_f. A more complete model for PTKE would include a separate equation for the PTKE dissipation rate (i.e., to replace $C_f k_f/\tau_p$). This would be useful, for example, to describe PTKE transport and decay behind a particle curtain (see Fig. 3) where $\alpha_p = 0$ and hence C_f is infinite.

The macroscale model for monodisperse gas–particle flow with isothermal particles consists of the eight balance equations given in Eqs. (12) to (19). Because the particle-phase VDF is approximated using a Maxwellian distribution, which is appropriate for collisional flows with $\alpha_p > 0.01$ (Fox, 2014), these equations appear in closed form. For smaller volume fractions, the number of balance equations can be increased to improve the prediction of the spatial fluxes (Fox and Laurent, 2022). For example, one can model the VDF as a Gaussian distribution and solve a balance equation for the velocity variance $\boldsymbol{\sigma}_p$ (i.e., the anisotropic Gaussian model Vié et al., 2015). In any case, whatever the level of closure used for the particle phase, it is very important to show that the macroscopic model is well-posed. Indeed, an ill-posed macroscale model will exhibit large oscillations in the conserved variables when solved numerically that have no physical relation with the solution to the mesoscale model from which the macroscale model was derived.

5.2. Is the macroscale model well-posed?

It is well known that the Archimedes' force, which couples the two phases through the continuous-phase pressure gradient, can lead to an ill-posed macroscale model (Lhuillier et al., 2013). This is especially true for large $\mathcal{Z}$ such as for bubbly flows. In comparison, without the coupling terms on the right-hand sides of Eqs. (12) to (19) the system has the form of the Euler equations for compressible flow, which are well-posed as long as the granular and gas temperatures are non-negative. In practice, the gas temperature will always be positive such that the speed of sound in the gas phase is large. However, due to energy dissipation, Θ_p can be very small and, thus, it will suffice to demonstrate that the system is well-posed when $\Theta_p = 0$. In the following, a brief review is provided on how to determine whether the 1-D version of the eight-equation macroscale model from Section 5.1 is well-posed. Because only terms with first-order derivatives are needed to determine whether the system is well-posed (Drew and Passman, 1999; Toro, 2009; Lhuillier et al., 2013), adding viscous stresses would play no role in the hyperbolicity of the macroscale model.

The conserved variables in the 1-D macroscale model form the state vector

$$\mathbf{X} = (\rho_f \alpha_f^\star, \rho_f \alpha_f^\star u_f, \rho_f \alpha_f^\star E_f, \rho_f \alpha_f^\star k_f, \rho_p \alpha_p, \rho_e \alpha_p^\star, \rho_e \alpha_p^\star u_p, \rho_e \alpha_p^\star K_p)^t.$$

Written in terms of the components of $\mathbf{X}$, the 1-D macroscale model for high-speed, monodisperse gas–particle flow, including added mass and pseudo-kinetic energy, is given in Table 1 where all terms not involving first-order derivatives have been discarded. In terms of the state vector, the 1-D macroscopic model can be written in the compact form of a hyperbolic conservation equation (Drew and Passman, 1999; Toro, 2009):

$$\partial_t \mathbf{X} + \partial_x \mathbf{F}(\mathbf{X}) + \mathbf{C}(\mathbf{X})\partial_x \mathbf{X} = \mathbf{S}(\mathbf{X}) \qquad \Longrightarrow \qquad \partial_t \mathbf{X} + \mathbf{J}(\mathbf{X})\partial_x \mathbf{X} = \mathbf{S}(\mathbf{X})$$

where the source terms $\mathbf{S}(\mathbf{X})$ do not depend on first-order space/time derivatives. In order for this system to be globally hyperbolic, the eight eigenvalues of the 8×8 Jacobian matrix $\mathbf{J} = \mathbf{C} + D\mathbf{F}/D\mathbf{X}$ must be real and distinct for all possible values of $\mathbf{X}$. Here, $\mathbf{C}$ arises due to the coupling terms, and includes the Archimedes' force. These terms will usually scale with $\mathcal{Z}$, i.e., they will be null when $\mathcal{Z} = 0$. The flux vector for the uncoupled system $\mathbf{F}$ and its Jacobian matrix $D\mathbf{F}/D\mathbf{X}$ with components $\partial F_j/\partial X_i$ depend on the value of $\mathbf{X}$. Thus, the eigenvalues of $\mathbf{J}$ depend also on $\mathbf{X}$ so that checking for hyperbolicity is not simple. Nonetheless, using symbolic software (Fox et al., 2020), it is straightforward to find the components of $\mathbf{J}$ starting from the macroscale model given in Table 1.

As shown elsewhere (Fox et al., 2020), thanks to the inclusion of added mass and the pfp pressure P_{pfp}, the 1-D macroscale model is well-posed for all physically relevant values of $\mathbf{X}$. Although it is not possible to compute the eigenvalues analytically, in most cases, two of the eight will be close to u_f and two others close to u_p. The other four eigenvalues will be close to the forms $u_f \pm c_f$ and $u_p \pm c_p$ where c_f and c_p are the speeds of sound in the gas/particle-phase, respectively. Note that these speeds will depend on α_p (and other variables) due to the coupling terms (i.e., to $\mathbf{C}$) in the macroscale model. Numerical algorithms for solving the macroscale model usually require knowledge of the eigenvalues (Houim and Oran, 2016; Boniou and Fox, 2023; Furfaro and Saurel, 2015; Liou, 2006; Toro, 2009; Boniou et al., 2023). Due to the large disparity between the speed of sound in the two phases, algorithms that treat the spatial fluxes $\mathbf{F}$ in each phase separately can be employed to improve the numerical accuracy (Houim and Oran, 2016). However, care must be taken to ensure that the coupling terms $\mathbf{C}\partial_x\mathbf{X}$ are treated in a consistent manner to avoid spurious solutions. Because the coupling terms will be very large, this is especially important when $\mathcal{Z} \gg 1$ (e.g., for bubbly flows).

5.3. Monodisperse macroscale model with constant fluid density

If the fluid is incompressible with ρ_f constant, then the identity $\alpha_p^\star + \alpha_f^\star = 1$ leads to the divergence-free constraint: $\partial_{\mathbf{x}} \cdot \mathbf{u}_v = 0$ where $\mathbf{u}_v = \alpha_p^\star \mathbf{u}_p + \alpha_f^\star \mathbf{u}_f$ is the mixture volumetric velocity. The latter provides a method to determine the fluid pressure. The resulting model for isothermal particles is given in Table 2 wherein only the drag and Archimedes' forces are included. In the momentum balances, the modified pressure p_f, defined to include k_f but without the hydrostatic component, is used. Written in this manner, the hydrostatic force (last term on right-hand side) depends on gravity $\mathbf{g}$ and the density ratio $\mathcal{Z}$. As noted above, p_f is found using the constraint $\partial_{\mathbf{x}} \cdot \mathbf{u}_v = 0$. All other terms are defined the same as for a compressible fluid. Because the pfp stresses do not affect the granular temperature, $\mathbf{P}_{pfp}$ does not appear in the fluctuating energy balances.

The eight-equation model in Table 2 is hyperbolic for all physical values $0 < \mathcal{Z} < \infty$ (Fox et al., 2020). Compared to the compressible case where the speed of sound in the fluid is finite, the speed of sound is infinite when ρ_f is constant. When checking the hyperbolicity of the system it suffices to consider only the two mass and two momentum balances with $\Theta_p = 0$. Then, for $\alpha_p < \alpha_{max}$, the hyperbolicity will be determined by the pfp stresses (Fox et al., 2020). The two-fluid model for this case has a separate pressure for each phase (i.e., p_f and the pfp pressure). This simplified four-equation system has two eigenvalues at $\pm\infty$, and the two other are real-valued. For $\alpha_p > \alpha_{max}$, the two eigenvalues are controlled by the frictional pressure P_{fr} and are real-valued as long as P_{fr} is a monotonically increasing function of α_p, which must always be true.

The added mass is important for $\mathcal{Z} > 0.01$, so it is of interest to examine the model behavior for very large $\mathcal{Z}$. In this case, the drag timescale becomes (using $\rho_e/\rho_p \to \mathcal{Z}\alpha_a/\alpha_p^\star$)

$$\mathcal{Z}\tau_p = \frac{\rho_e}{\rho_p}\frac{4d_p^2}{3\nu_f C_D Re_p} \quad \to \quad \mathcal{Z}\frac{\alpha_a}{\alpha_p^\star}\hat{\tau}_p,$$

Table 1
Eight-equation macroscale model for 1-D gas–particle flow written in terms of the components of the state vector $\mathbf{X}$. The source terms not involving temporal or spatial gradients are set to zero for clarity.

Gas-phase equations:

$$\partial_t X_1 + \partial_x F_1 = 0$$
$$\partial_t X_2 + \partial_x F_2 + (X_6/\rho_e)\partial_x \hat{p}_f = 0$$
$$\partial_t X_3 + \partial_x F_3 - (X_7/\rho_e)\partial_x \hat{p}_f + P_{pfp}\partial_x(X_7/X_6) = 0$$
$$\partial_t X_4 + \partial_x F_4 + \frac{2}{3}X_4\partial_x(X_2/X_1) = 0$$

Particle-phase equations:

$$\partial_t X_5 + \partial_x F_5 = 0$$
$$\partial_t X_6 + \partial_x F_6 = 0$$
$$\partial_t X_7 + \partial_x F_7 - (X_6/\rho_e)\partial_x \hat{p}_f = 0$$
$$\partial_t X_8 + \partial_x F_8 + (X_7/\rho_e)\partial_x \hat{p}_f - P_{pfp}\partial_x(X_7/X_6) = 0$$

where the spatial fluxes are

$$F_1 = X_2 \qquad F_5 = X_5 X_7/X_6$$
$$F_2 = X_2^2/X_1 + \hat{p}_f \qquad F_6 = X_7$$
$$F_3 = X_2 X_3/X_1 + X_2\hat{p}_f/\rho_f + X_7\hat{p}_f/\rho_e \qquad F_7 = X_7^2/X_6 + X_6(\Theta_p + P_p) + P_{pfp}$$
$$F_4 = X_2 X_4/X_1 \qquad F_8 = X_7 X_8/X_6 + X_7(\Theta_p + P_p) + (X_7/X_6)P_{pfp}$$

The remaining variables that depend on $\mathbf{X}$ are defined by

$$\kappa = \frac{X_6 - X_5}{X_1} \qquad \alpha_p = X_5/\rho_p \qquad \alpha_f = 1 - \alpha_p \qquad \alpha_a = \frac{\kappa}{1+\kappa}\alpha_f \qquad \alpha_p^\star = \alpha_p + \alpha_a \qquad \alpha_f^\star = \alpha_f - \alpha_a$$
$$P_{pfp} = C_{pfp}\rho_f\alpha_p^\star u_{pf}^2 \qquad u_{pf} = X_7/X_6 - X_2/X_1 \qquad \rho_f = X_1/\alpha_f^\star \qquad \rho_e = (X_5 + \rho_f\alpha_a)/\alpha_p^\star$$
$$\hat{p}_f = p_f + \frac{2}{3}\rho_f X_4/X_1 \qquad p_f = (\gamma_f - 1)\rho_f e_f \qquad e_f = X_3/X_1 - \frac{1}{2}(X_2/X_1)^2 - X_4/X_1$$
$$P_p = 2(1+e_c)\alpha_p g_0 \Theta_p + \frac{p_{fr}\alpha_p g_0 h_{fr}}{X_6} \qquad \Theta_p = \frac{2}{3}X_8/X_6 - \frac{1}{3}(X_7/X_6)^2$$

with constants ρ_p, γ_f, C_{pfp} and p_{fr}.

Table 2
Eight-equation model for constant-density, inviscid, fluid–particle flow with added mass, internal energy, and PTKE. The particles are assumed to be isothermal and p_f is the modified fluid pressure. The mass balances use β in place of $\alpha_p = \alpha_p^\star - \alpha_a$ where $\alpha_a/\alpha_p^\star = (\beta - 1)/(\beta - 1 + \mathcal{Z})$. The momentum and internal energy balances are normalized by dividing them by their material densities, and thus the model depends on the density ratio $\mathcal{Z} = \rho_f/\rho_p$ and not the individual densities. Written in this form, the three limiting cases: $\mathcal{Z} = 0, 1, \infty$ can be easily identified. The slip velocities are $\mathbf{u}_{fp} = -\mathbf{u}_{pf} = \mathbf{u}_f - \mathbf{u}_p$. The added-mass factor is $\beta = 1 + \mathcal{Z}\alpha_a/\alpha_p$ such that $\mathcal{Z}\tau_p$ is finite for $\mathcal{Z} = 0$. In the complete model, fluid-phase viscosity and thermal conductivity will be included in the momentum and internal-energy balances, respectively. Note that these processes will be enhanced by PTKE (Zhou et al., 2024); hence, their coefficients will depend on $\alpha_f^\star d_p k_f^{1/2}$, and it is for this reason that a balance for k_f is required.

Added-mass factor:

$$\partial_t\beta + \mathbf{u}_p\cdot\partial_{\mathbf{x}}\beta = \frac{\mathcal{Z}}{\alpha_p}\frac{1}{\rho_f}S_a = \frac{1}{\tau_a}(\beta^\star - \beta)$$

Mass balances:

$$\partial_t\alpha_f^\star + \partial_{\mathbf{x}}\cdot(\alpha_f^\star\mathbf{u}_f) = -\frac{1}{\rho_f}S_a$$
$$\partial_t\alpha_p^\star + \partial_{\mathbf{x}}\cdot(\alpha_p^\star\mathbf{u}_p) = \frac{1}{\rho_f}S_a$$

Momentum balances:

$$\partial_t(\alpha_f^\star\mathbf{u}_f) + \partial_{\mathbf{x}}\cdot(\alpha_f^\star\mathbf{u}_f\mathbf{u}_f) + \alpha_f^\star\partial_{\mathbf{x}}p_f = \tfrac{\beta\alpha_p}{\mathcal{Z}\tau_p}\mathbf{u}_{pf} - \tfrac{1}{\rho_f}\mathbf{S}_a - \tfrac{1-\mathcal{Z}}{\mathcal{Z}}\alpha_f^\star\alpha_p\mathbf{g}$$
$$\partial_t(\beta\alpha_p\mathbf{u}_p) + \partial_{\mathbf{x}}\cdot\left[\beta\alpha_p(\mathbf{u}_p\mathbf{u}_p + \Theta_p\mathbf{I} + P_p\mathbf{I}) + \tfrac{1}{\rho_p}\mathbf{P}_{pfp}\right] + \mathcal{Z}\alpha_p^\star\partial_{\mathbf{x}}p_f = \tfrac{\beta\alpha_p}{\tau_p}\mathbf{u}_{fp} + \tfrac{1}{\rho_p}\mathbf{S}_a + (1-\mathcal{Z})\alpha_f^\star\alpha_p\mathbf{g}$$

Fluctuating kinetic-energy balances:

$$\partial_t(\alpha_f^\star k_f) + \partial_{\mathbf{x}}\cdot(\alpha_f^\star k_f\mathbf{u}_f) + \frac{2}{3}\alpha_f^\star k_f\partial_{\mathbf{x}}\cdot\mathbf{u}_f = \frac{\beta\alpha_p}{\mathcal{Z}\tau_p}[3a\Theta_p - 2(1-a)k_f + u_{pf}^2 - C_f k_f]$$
$$\partial_t\left(\beta\alpha_p\frac{3}{2}\Theta_p\right) + \partial_{\mathbf{x}}\cdot\left(\beta\alpha_p\frac{3}{2}\Theta_p\mathbf{u}_p\right) + \beta\alpha_p(\Theta_p + P_c)\partial_{\mathbf{x}}\cdot\mathbf{u}_p = -\beta\alpha_p\left[\frac{1}{\tau_p}[3a\Theta_p - 2(1-a)k_f] + \frac{1}{\tau_p}(1-e_c^2)\Theta_p + \frac{1}{\tau_{fr}}\Theta_p\right]$$

Internal-energy balance for fluid (e_p is assumed constant):

$$\partial_t(\alpha_f^\star e_f) + \partial_{\mathbf{x}}\cdot(\alpha_f^\star e_f\mathbf{u}_f) = \frac{\beta\alpha_p}{\mathcal{Z}}\left[\frac{1}{\tau_p}C_f k_f + \frac{1}{\tau_p}(1-e_c^2)\Theta_p + \frac{1}{\tau_{fr}}\Theta_p\right]$$

which is linear in $\mathcal{Z}$ such that

$$\frac{\beta\alpha_p}{\mathcal{Z}\tau_p} \quad\to\quad \frac{\alpha_p}{\hat{\tau}_p}.$$

Using this result, the momentum balances for $\mathcal{Z}\to\infty$ reduce to

$$\partial_t(\alpha_f^\star\mathbf{u}_f) + \partial_{\mathbf{x}}\cdot(\alpha_f^\star\mathbf{u}_f\mathbf{u}_f) + \alpha_f^\star\partial_{\mathbf{x}}p_f = \frac{\alpha_p}{\hat{\tau}_p}\mathbf{u}_{pf} - \frac{1}{\rho_f}\mathbf{S}_a + \alpha_f^\star\alpha_p\mathbf{g}$$

and

$$\partial_t(\alpha_a\mathbf{u}_p) + \partial_{\mathbf{x}}\cdot\left[\alpha_a(\mathbf{u}_p\mathbf{u}_p + \Theta_p\mathbf{I} + P_p\mathbf{I}) + \frac{1}{\rho_f}\mathbf{P}_{pfp}\right] + \alpha_p^\star\partial_{\mathbf{x}}p_f = \frac{\alpha_p}{\hat{\tau}_p}\mathbf{u}_{fp} + \frac{1}{\rho_f}\mathbf{S}_a - \alpha_f^\star\alpha_p\mathbf{g}$$

where the pfp stresses for spherical bubbles in potential flow are

$$\mathbf{P}_{pfp} = C_{pfp}\rho_f\alpha_p^\star\left(\frac{1}{5}u_{pf}^2\mathbf{I} + \frac{2}{5}\mathbf{u}_{pf}\otimes\mathbf{u}_{pf}\right)$$

with (using the added-mass notation with the pfp-stress coefficient derived for potential flow in Bulthuis et al., 1995; Spelt and Sangani, 1998)

$$C_{pfp}(\alpha_p) = \frac{1}{2}\frac{\alpha_p^\star\alpha_f^\star}{\alpha_p\alpha_f},$$

which is valid for $\alpha_p < 0.5$. Note that $C_{pfp}(0) = 3/4$ and $C_{pfp}(0.5) = 3/8$ for bubbly flow. For $\alpha_p > 0.5$, the constant $C_{pfp} = 3/8$ can be used. The added-mass momentum source term $\mathbf{S}_a/\rho_f$ is proportional to S_a/ρ_f, which reduces to

$$\frac{1}{\rho_f}S_a \quad\rightarrow\quad \frac{\alpha_p}{\hat{\tau}_a\beta}(\beta^\star - \beta)$$

with $\hat{\tau}_a \propto \hat{\tau}_p$. The forces on the right-hand sides of the momentum balances therefore all scale with α_p.

For the spatially homogeneous case, the modified pressure p_f is constant since the hydrostatic contribution is included in the gravity force, and the other terms on the right-hand side determine $\mathbf{u}_{pf}$. Otherwise, for unsteady, spatially inhomogeneous flows, the effective mass of a particle is $\rho_f\alpha_a$, i.e., the added mass, and hence the rate of acceleration of α_a scales with $\alpha_p/\alpha_a \approx 2$. Thus, accounting for the added mass will lower the acceleration compared to the case where it is neglected. The numerical treatment of the force terms is therefore simpler due to the larger characteristic timescale. Finally, it should be noted that in liquid–particle flows the viscosity of the fluid phase can be significant in real applications. Here, the viscous stresses have been neglected for simplicity, but can be added to the fluid-phase model without modifying the overall mathematical structure of the macroscale model in Table 2.

5.4. Extension of the macroscale model to polydisperse particles

With the macroscale model, one does not solve for the NDF, but rather for its moments (Fox et al., 2024). For the polydisperse case, one must decide which moments are needed to find a closed system of equations that is well-posed. This choice will depend on how the unknown NDF is approximated by its moments. For example, with quadrature-based moment methods (QBMM) (Marchisio and Fox, 2013), the required moments are the size moments $\mathcal{M}_k$ for $k \geq 0$, which provide a discrete representation of the size distribution function:

$$n(\xi) = \sum_{i=1}^{N} w_i\delta(\xi - \xi_i)$$

with positive weights $w_i(t,\mathbf{x})$ and distinct abscissae $\xi_i(t,\mathbf{x})$. With the QBMM approximation, the size moments are related to the weights and abscissae by

$$M_k = \int \xi^k n(\xi)\,\mathrm{d}\xi = \sum_{i=1}^{N} w_i\xi_i^k$$

and

$$\mathcal{M}_{k+1} = \int \xi^k f(\mathbf{u},\xi)\,\mathrm{d}\mathbf{u}\,\mathrm{d}\xi = \sum_{i=1}^{N}\beta w_i\xi_i^{k+1} = \sum_{i=1}^{N} n_i\xi_i^{k+1}.$$

The number of size moments required depends on the complexity of the shape of the size NDF. For example, a uni-modal NDF with a small variance requires less moments than a multi-modal NDF with large variance. In general, the number of equations in the particle-phase macroscale model scales with N. Thus, choosing a larger N increases the accuracy of the QBMM approximation, but also increases the cost of solving the macroscale model numerically.

Hereinafter, for simplicity, the size moments $(\mathcal{M}_0,\mathcal{M}_1,\mathcal{M}_2)$ are used to approximate the particle size distribution with $N = 3$. Knowledge of these moments (which are equivalent to the number concentration and the mean and variance of the size NDF), along with the generalized quadrature method of moments (GQMOM) (Fox et al., 2023a), allows one to find the weights n_i and abscissas ξ_i. Note that β acts as a scaling factor for all moments; thus, the abscissae will be the same when found with the moments (M_0, M_1, M_2). In general, QBMM provide an accurate representation of the NDF using the minimal number of moments (Marchisio and Fox, 2013). Moreover, the resulting form of the macroscale balances makes it possible to demonstrate that the macroscale model for the selected moments is well-posed. More details can be found in Fox et al. (2024).

Next, in order to find the velocity and kinetic energy corresponding to a particle with size ξ_i, the conditional quadrature method of moments (CQMOM) is employed (Yuan and Fox, 2011). For this purpose, the following joint moments are required (Fox et al., 2024):

- Size–velocity moments: $\boldsymbol{\mathcal{U}}_{k+1} = \int \xi^k\mathbf{u}f\,\mathrm{d}\mathbf{u}$ where $\boldsymbol{\mathcal{U}}_1 = \rho_e\alpha_p^\star\mathbf{u}_p$ is the particle-phase momentum.
- Size-kinetic-energy moments: $\mathcal{K}_{k+1} = \int \xi^k\frac{1}{2}u^2 f\,\mathrm{d}\mathbf{u}$ where $\mathcal{K}_1 = \rho_e\alpha_p^\star K_p$ is the particle-phase kinetic energy.

The nine conserved moments used in the macroscale model are then $(\mathcal{M}_0,\mathcal{M}_1,\mathcal{M}_2)$, $(\boldsymbol{\mathcal{U}}_0,\boldsymbol{\mathcal{U}}_1,\boldsymbol{\mathcal{U}}_2)$, and $(\mathcal{K}_0,\mathcal{K}_1,\mathcal{K}_2)$. Thus, starting from the kinetic equation in Eq. (11), the unclosed balance equations for these nine conserved variables must be derived (Fox et al., 2024). Then, to close these balance equations, the joint size–velocity NDF is approximated as

$$f(\mathbf{u},\xi) = \sum_{i=1}^{3} n_i p(\mathbf{u}|\xi_i)\,\xi\,\delta(\xi - \xi_i) \tag{20}$$

where $p(\mathbf{u}|\xi_i)$ is the conditional probability distribution for the velocity of particles with size ξ_i, mean velocity $\mathbb{U}_i = \int \mathbf{u}p(\mathbf{u}|\xi_i)\,\mathrm{d}\mathbf{u}$ and kinetic energy $\mathbb{K}_i = \int \frac{1}{2}u^2 p(\mathbf{u}|\xi_i)\,\mathrm{d}\mathbf{u}$. In order to be consistent with the closure used for the VDF in the monodisperse case (i.e., only depending on the mean velocity and kinetic energy), it is assumed that $p(\mathbf{u}|\xi_i)$ is Maxwellian for each ξ_i. This closure should be adequate for collisional and/or low-Stokes-number flows where the kinetic energy of the particles is close local equilibrium (Fox et al., 2024). Otherwise, at the next level of closure, $p(\mathbf{u}|\xi_i)$ can be modeled by a Gaussian distribution with conditional velocity covariance $\boldsymbol{\sigma}_p(\xi_i)$. In that case, the three moments $(\mathcal{K}_0,\mathcal{K}_1,\mathcal{K}_2)$ would be replaced the symmetric second-order tensors $\boldsymbol{\mathcal{U}}_k^2 = \int \xi^k\mathbf{u}\otimes\mathbf{u}f\,\mathrm{d}\mathbf{u}$ for $k = 0,1,2$. These moments are related by $2\mathcal{K}_k = trace(\boldsymbol{\mathcal{U}}_k^2)$. Otherwise, if solutions to the polydisperse kinetic equation exhibit non-equilibrium behavior, even higher-order velocity moments can be used as shown in Fox and Laurent (2022). The transport equations for such moments are given in Fox et al. (2024).

In summary, using QBMM the relationships between the nine conserved moments in the mesoscale model and the conditional moments are as follows:

- Given $(\mathcal{M}_0,\mathcal{M}_1,\mathcal{M}_2)$, the weights and abscissae $(n_1,\xi_1),(n_2,\xi_2),(n_3,\xi_3)$ are found using GQMOM such that
$$\mathcal{M}_k = n_1\xi_1^k + n_2\xi_2^k + n_3\xi_3^k \qquad \text{for } k = 0,1,2.$$
- Using CQMOM, the three size-conditioned velocities $(\mathbb{U}_1,\mathbb{U}_2,\mathbb{U}_3)$ are found from
$$\boldsymbol{\mathcal{U}}_k = n_1\xi_1^k\mathbb{U}_1 + n_2\xi_2^k\mathbb{U}_2 + n_3\xi_3^k\mathbb{U}_3 \qquad \text{for } k = 0,1,2.$$
- Using CQMOM, the three size-conditioned kinetic energies $(\mathbb{K}_1,\mathbb{K}_2,\mathbb{K}_3)$ are found from
$$\mathcal{K}_k = n_1\xi_1^k\mathbb{K}_1 + n_2\xi_2^k\mathbb{K}_2 + n_3\xi_3^k\mathbb{K}_3 \qquad \text{for } k = 0,1,2.$$

Notice that with CQMOM, a linear system must be solved for the six unknown conditional variables:

$$\begin{bmatrix} n_1 & n_2 & n_3 \\ n_1\xi_1 & n_2\xi_2 & n_3\xi_3 \\ n_1\xi_1^2 & n_2\xi_2^2 & n_3\xi_3^2 \end{bmatrix}\begin{bmatrix} \mathbb{U}_1 & \mathbb{K}_1 \\ \mathbb{U}_2 & \mathbb{K}_2 \\ \mathbb{U}_3 & \mathbb{K}_3 \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mathcal{U}}_0 & \mathcal{K}_0 \\ \boldsymbol{\mathcal{U}}_1 & \mathcal{K}_1 \\ \boldsymbol{\mathcal{U}}_2 & \mathcal{K}_2 \end{bmatrix}.$$

This system is non-singular because the weights n_i found from GQMOM are positive and the abscissae ξ_i are distinct. The conditional moments $\mathbb{K}_i$ must be positive and are realizable provided that $2\mathbb{K}_i \geq \boldsymbol{\mathcal{U}}_i \cdot \boldsymbol{\mathcal{U}}_i$, which corresponds to the conditional granular temperature Θ_i being non-negative. If the solution to the linear system is not realizable, a correction algorithm can be applied (Fox et al., 2024).

The final step is to derive the nine transport equations for the particle phase in the macroscale model. Starting from the polydisperse kinetic model in Eq. (11), these equations are as follows (Fox et al., 2024):

- For the size moments with $k = 0, 1, 2$:

$$\partial_t \mathcal{M}_k + \partial_{\mathbf{x}} \cdot \boldsymbol{\mathcal{U}}_k = S_k^0. \tag{21}$$

- For the size–velocity moments with $k = 0, 1, 2$:

$$\partial_t \boldsymbol{\mathcal{U}}_k + \partial_{\mathbf{x}} \cdot (\boldsymbol{\mathcal{U}}_k^2 + \mathcal{M}_k P_p \mathbf{I}) + \frac{\mathcal{M}_k}{\mathcal{M}_1} \partial_{\mathbf{x}} \cdot \mathbf{P}_{pfp} = -\mathcal{A}_k^1 - \alpha_p^\star \frac{\mathcal{M}_k}{\mathcal{M}_1} \partial_{\mathbf{x}} \hat{p}_f + \mathcal{C}_k^1 + \mathcal{F}_k^1 + \mathcal{S}_k^1 + \mathcal{M}_k \mathbf{g}. \tag{22}$$

- For the size-kinetic-energy moments with $k = 0, 1, 2$:

$$\partial_t \mathcal{K}_k + \partial_{\mathbf{x}} \cdot \left(\boldsymbol{\mathcal{U}}_k^K + \boldsymbol{\mathcal{U}}_k^1 P_p + \frac{\boldsymbol{\mathcal{U}}_k^1}{\mathcal{M}_1} \cdot \mathbf{P}_{pfp} \right) = \mathbf{P}_{pfp} : \partial_{\mathbf{x}} \left(\frac{\boldsymbol{\mathcal{U}}_k^1}{\mathcal{M}_1} \right) - \mathcal{A}_k^2 + (1-a)\mathcal{B}_k^2 - \alpha_p^\star \frac{\boldsymbol{\mathcal{U}}_k^1}{\mathcal{M}_1} \cdot \partial_{\mathbf{x}} \hat{p}_f + \mathcal{C}_k^2 + \mathcal{F}_k^2 + \mathcal{S}_k^2 + \boldsymbol{\mathcal{U}}_k^1 \cdot \mathbf{g}. \tag{23}$$

The added-mass source terms on the right-hand sides are defined by

$$S_k^0 = C_\xi \mathcal{M}_k, \quad S_k^1 = C_\xi \mathcal{M}_k \mathbf{u}_f + (C_\xi - C_u)\boldsymbol{\mathcal{U}}_k^1,$$
$$S_k^2 = C_\xi \mathcal{M}_k K_f + (C_\xi - C_u)\mathcal{K}_k,$$

and are in closed form (i.e., they depend on the conserved moments). The same is true for the Archimedes' force, the pfp-stress terms, and gravity. Otherwise, all of the unclosed terms are approximated by using Eq. (20).

The spatial fluxes of the conserved moments are closed by using the size-conditioned velocities/energies (Fox et al., 2024). For example, with the Maxwellian closure for the conditional velocity NDF, the following closures for the spatial fluxes are employed:

$$\boldsymbol{\mathcal{U}}_k^2 = n_1 \xi_1^k (\mathbb{U}_1 \otimes \mathbb{U}_1 + \Theta_1 \mathbf{I}) + n_2 \xi_2^k (\mathbb{U}_2 \otimes \mathbb{U}_2 + \Theta_2 \mathbf{I}) + n_3 \xi_3^k (\mathbb{U}_3 \otimes \mathbb{U}_3 + \Theta_3 \mathbf{I})$$

and

$$\boldsymbol{\mathcal{U}}_k^K = n_1 \xi_1^k \mathbb{U}_1 (\mathbb{K}_1 + \Theta_1) + n_2 \xi_2^k \mathbb{U}_2 (\mathbb{K}_2 + \Theta_2) + n_3 \xi_3^k \mathbb{U}_3 (\mathbb{K}_3 + \Theta_3).$$

The contribution of the collisional and frictional pressures P_p has been averaged over the size distribution using QBMM, and $\mathbf{P}_{pfp}$ is defined the same way as in the monodisperse case. From a numerical point of view, it is possible to split the spatial fluxes and the source/coupling terms (Fox et al., 2024). Then, the most efficient numerical routines can be used for each part. For example, the source terms such as collisions (Andries et al., 2002) can be most easily handled with the conditional variables since the weights and abscissae do not change during a particle–particle collision (Fox et al., 2024). In general, splitting the different terms makes it easier to ensure the conservation of mass, momentum and energy, as well as to keep the conserved and conditional moments realizable (Passalacqua et al., 2018; Fox et al., 2024).

The drag, collision, and friction source terms are defined in terms of the size-conditioned velocities and kinetic energies (Fox et al., 2024). For example, in the momentum balances, the drag term is

$$\mathcal{A}_k^1 = n_1 \xi_1^k \frac{1}{\tau_p(\xi_1)} (\mathbb{U}_1 - \mathbf{u}_f) + n_2 \xi_2^k \frac{1}{\tau_p(\xi_2)} (\mathbb{U}_2 - \mathbf{u}_f) + n_3 \xi_3^k \frac{1}{\tau_p(\xi_3)} (\mathbb{U}_3 - \mathbf{u}_f)$$

where $\tau_p(\xi)$ is the size-dependent drag timescale. In general, all terms are analogous to the monodisperse case. In particular, for $k = 1$ the balance equations represent the particle-phase mass, momentum and kinetic-energy balances that are coupled to the fluid phase. Furthermore, the four fluid-phase equations are the same as for the monodisperse case, differing only in the coupling terms shared with the particle-phase mass, momentum and kinetic-energy balances.

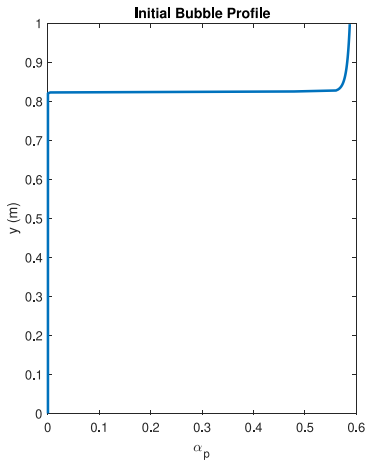


Fig. 13. Initial distribution of the volume fraction of particles.

In summary, the macroscale model for polydisperse particles with $N = 3$ abscissae has a total of fourteen equations, ten for the particle phase (including the mass balance for $\rho_p \alpha_p$) and four for the fluid phase. In the following, some example applications of the macroscale model to different monodisperse and polydisperse two-phase flows are provided.

6. Example applications of macroscale models

In this section a few example applications of the macroscale models described in the previous section are provided. Each example is selected to highlight a particular aspect of the flow physics such as the frictional pressure and the pfp stresses. Results for both the monodisperse and the polydisperse models are presented. Because it is assumed that the flows are spatially homogeneous in directions normal to the flow direction, all the example applications can be simulated using the 1-D macroscale model. This represents a considerable computational cost reduction compared to micro/mesoscale DNS that requires a 3-D domain. Thus, due to this simplification, the fluid-phase viscous terms play no role in the simulation results. Nevertheless, in many cases, such as gravity-driven flows, 3-D structures arise spontaneously and require the full 3-D macroscale model, including the viscous terms that have been neglected in this work.

6.1. Creaming of light particles in water

Creaming is the opposite of sedimentation in that light particles are allowed to rise under gravity. For convenience, here the direction of gravity is switched to point upward so that light particles will move downward. The parameters used to solve the incompressible model with monodisperse particles are $\rho_f = 1000$ kg/m^3, $\rho_p = 1$ kg/m^3, and $d_p = 1$ mm. The container is closed and the initial conditions for α_p are shown in Fig. 13. These initial conditions were computed by letting gravity point downward and running to steady state. Note that the maximum of α_p at the top is approximately 0.59, and is fixed by a force balance between buoyancy and the frictional pressure, which resists compaction of the particle phase. Thus, the monodisperse particles form a 'foam' at the top of the container with density $\alpha_p < \alpha_{max} = 0.63$ controlled by P_{fr}.

Starting from the initial condition in Fig. 13, the upward pointing gravity causes the particles to move downward as shown in Fig. 14.

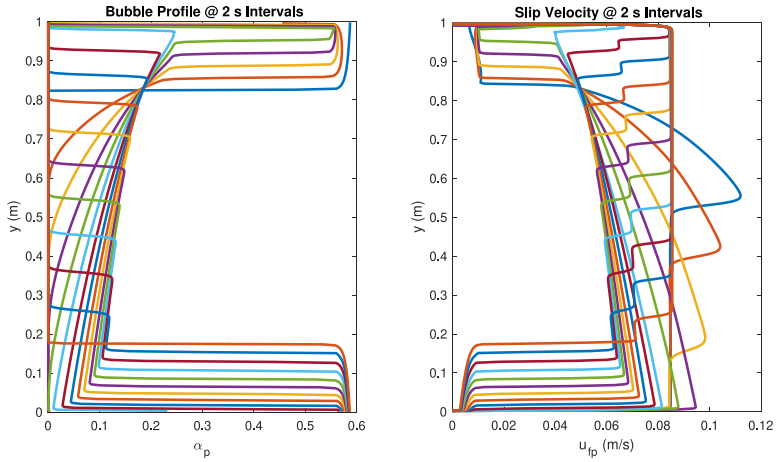


Fig. 14. Creaming of light particles in water. Blue curve at top is initial conditions in Fig. 13. Orange curve at bottom is final steady-state condition. The slip velocity depends on the drag coefficient, which depends on the local value of α_p. The 'foam' at the top disappears near 10 s (light-blue curve). At long times, the 'foam' reappears at the bottom. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In this example, the Richardson–Zaki correlation for the drag coefficient (Richardson and Zaki, 1954) is used, which has a strong dependence on the local value of α_p. Thus, the slip velocity is highest for small α_p and lowest for large α_p. This can be clearly observed in the 'bubble profile' where particles at the bottom of the 'foam' move down faster, generating an upflow of the liquid. This upflow can be observed in $\mathbf{u}_{fp}$, which is always positive. Notice that the slip velocity at the top jumps suddenly between 8–10 s (green and light blue curves), which corresponds to the 'foam' phase disappearing. After this time, all particles are moving downward with a sharp upper edge, above which there are no particles. In other words, there is a clear separation between the clear liquid at the top and the particle phase at the bottom, which corresponds to experimental observations of creaming. At the bottom of the container, the 'foam' is quickly reestablished when sufficient particles have arrived at the bottom. Because of the frictional pressure, the maximum value of α_p reaches again to approximately 0.59. At the end of the simulation, the 'foam' at the bottom is a mirror image of the initial conditions.

In summary, the physics of particle creaming are reproduced by the macroscale model. The frictional pressure and gravity fix the final state of the 'foam' phase. Here, because monodisperse particles without coalescence are considered, the 'foam' is stable. In terms of the physical models at play during the transition process, the pfp stresses are very important because they ensure that the system remains hyperbolic, but they also affect the creaming velocity since they push particles apart when the slip velocity is high. In comparison, the value of Θ_p, and hence P_p, is small compared to P_{pfp}, which is expected considering the density ratio $\mathcal{Z} = 1000$. From this example, the model for P_{fr} plays an important role in fixing the volume fraction in the 'foam'. In most two-fluid models used for bubbly flows, P_{fr} is neglected, even though the bubbles are assumed to be monodisperse. Without the frictional pressure, α_p will continue to rise until it reaches unity, which can cause the simulation code to fail.[5] Such behavior may be appropriate for bubbles that immediately coalesce, but certainly not for non-coalescing bubbles that form a foam. In this example, the model in Section 4.1 is used for the frictional pressure, which was designed for non-deformable (stiff) particles. For the case of deformable bubbles, the functional form of $P_{fr}(\alpha_p)$ would need to be modified to allow for α_p in the foam to reach the correct physical value.

[5] Most production codes for the two-fluid model are designed to handle the limiting cases of $\alpha_p = 0$ and 1 without failing. To do so, the model equations must augmented with special conditions to handle these limits. This occurs, for example, in bubble columns at the gas–liquid interface at the top of the liquid phase.

6.2. Sedimentation of heavy particles in water

In this example, the same parameters from the previous example are used except for ρ_p, which is increased such that $\mathcal{Z} = 0.5$. The initial conditions are again shown in Fig. 13 and gravity is pointed downward relative to the y axis. As seen in Fig. 15, the temporal behavior of the volume fraction and slip velocity is very close to that observed in Fig. 14. This is due in fact to the added-mass factor that makes $m_p^\star$ approximately the same in both examples. Thus, the physical description of the important terms in the model are the same as for creaming, particularly concerning the roles of the pfp stresses and the frictional pressure. In Fig. 16 the time evolution of the PTKE, granular temperature and pfp pressure are plotted at the same times as in Fig. 15. The pfp pressure is proportional to $\mathcal{Z}\rho_f\alpha_p u_{pf}^2$, so that dividing it by ρ_f puts all quantities on the same scale.

Note that at the beginning and end of the simulation where $\mathbf{u}_p$ is null, all of the quantities in Fig. 16 are null. Thus, initially they all increase to a dynamic steady state, before collapsing at the end of simulation when all particles reach the bottom of the container. It can be observed that all quantities in Fig. 16 have the same profile with the largest values in the regions where the slip velocity is highest. Comparing their maximum magnitudes, it can be seen that the PTKE is larger than Θ_p. This is expected due to the relatively small Stokes number of the particles (i.e., small τ_p), which causes significant dissipation of granular energy. On the other hand, PTKE and P_{pfp}/ρ_f have the same order of magnitude. Note that the maximum pfp pressure is approximately 0.38 Pa, which is small compared p_f, but it is important to recall that the forces on the particles are proportional to the pressure gradients. Indeed, the gradient of P_{pfp} is large enough to ensure that the macroscopic model is hyperbolic (which is not the case for the kinetic pressure $P_p \propto \alpha_p\Theta_p$).

6.3. Shock traversing polydisperse particle curtain

This application involves a shock wave interacting with a thin curtain of polydisperse particles (Daniel and Wagner, 2022; Wagner et al., 2023; Capecelatro and Wagner, 2024; Boniou and Fox, 2023). The experimental set up is shown in Fig. 17. A shock enters the domain from the left and eventually traverses the particle curtain. Before the shock arrives at the curtain, $\mathbf{u}_p$ is null in the curtain, but rapidly increases as the shock passes. At the shock, there is a large jump in p_f that produces a rapid acceleration of the particles on the front side of the curtain. The initial conditions for gas density and temperature across the shock and the particle size distribution are as follows:

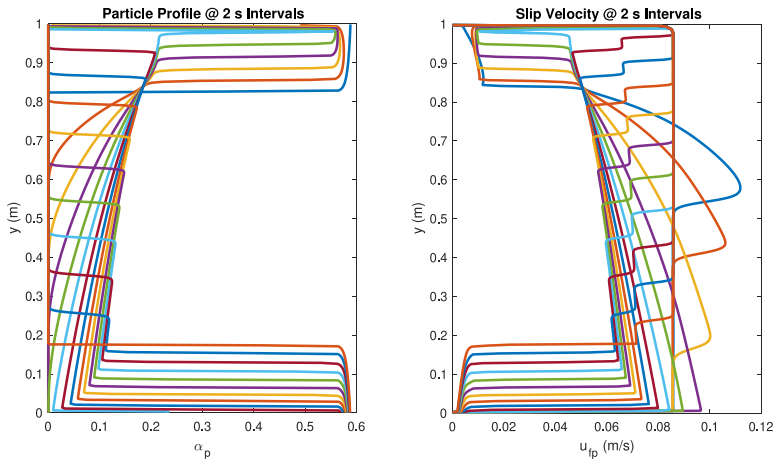


Fig. 15. Sedimentation of heavy particles in water. Due to the added-mass factor, the physical behavior is very close to that seen for creaming in Fig. 14. In both cases, at the end of the simulation the particles pack together at the bottom of the container. The final value of α_p is fixed by gravity and the frictional pressure.

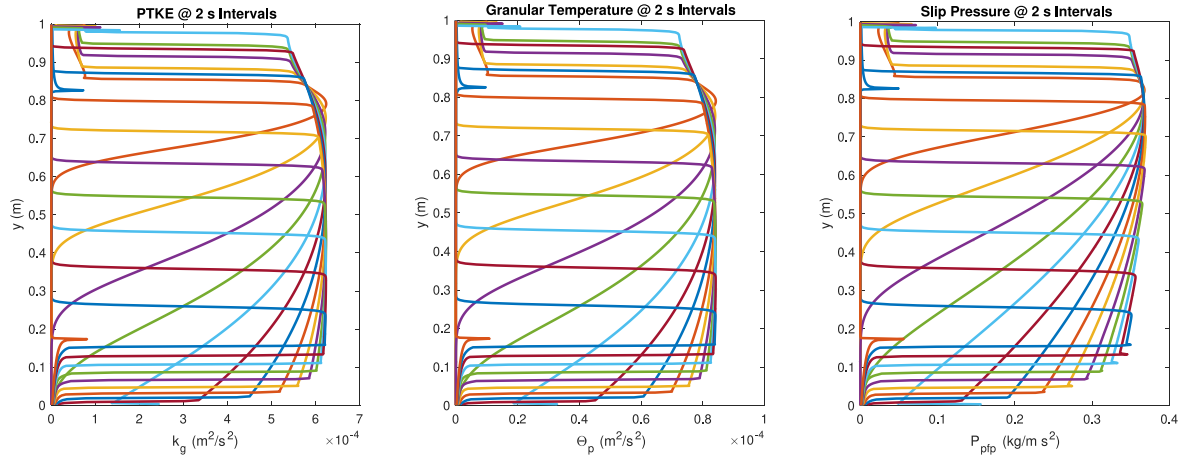


Fig. 16. Sedimentation of heavy particles in water. The time evolution of the PTKE (left), Θ_p (middle) and P_{pfp} (right). Dividing P_{pfp} by ρ_f (i.e., 1000) puts all quantities on the same scale. Colors correspond to the same times as in Fig. 15. All quantities are zero at the beginning and end of the simulation where $\mathbf{u}_{fp} = 0$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

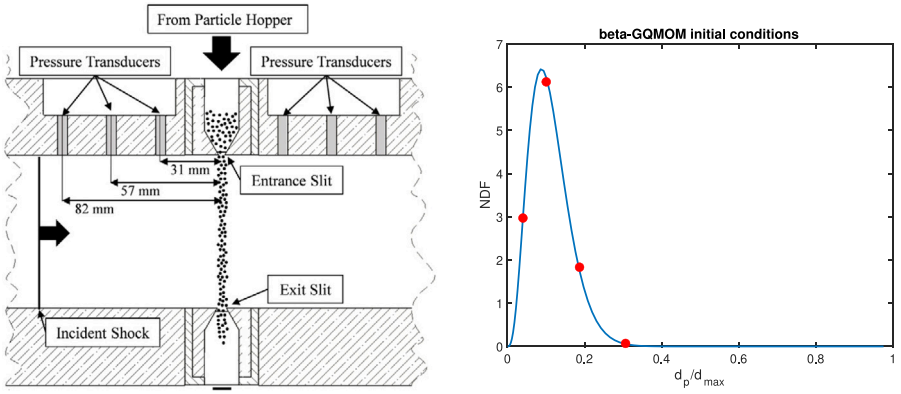


Fig. 17. Shock traversing polydisperse particle curtain (Daniel and Wagner, 2022). Left: experimental set up with shock tube and particle curtain. Right: Particle size distribution used for simulations. Red dots indicate the GQMOM weights and abscissae in the initial particle curtain. Image on left courtesy of Kyle Daniel and Justin Wagner.

- Left side: $\rho_f = 2.48$ kg/m^3, $T_f = 429$ K, $u_f = 0$.
- Right side: $\rho_f = 1.18$ kg/m^3, $T_f = T_p = 300$ K, $u_f = u_p = 0$, $\rho_p = 2520$ kg/m^3.
- Initial particle size: Beta distribution with $d_{max} = 10^{-4}$ m.

The initial particle size distribution is shown in Fig. 17, along with the weights and scaled abscissae found from GQMOM with $N = 4$ characteristic diameters. The particles are assumed to be spherical with the same density ρ_p. In this example, $\mathcal{Z}$ is small, but the slip velocity after the shock passes the curtain is very high, which results in large

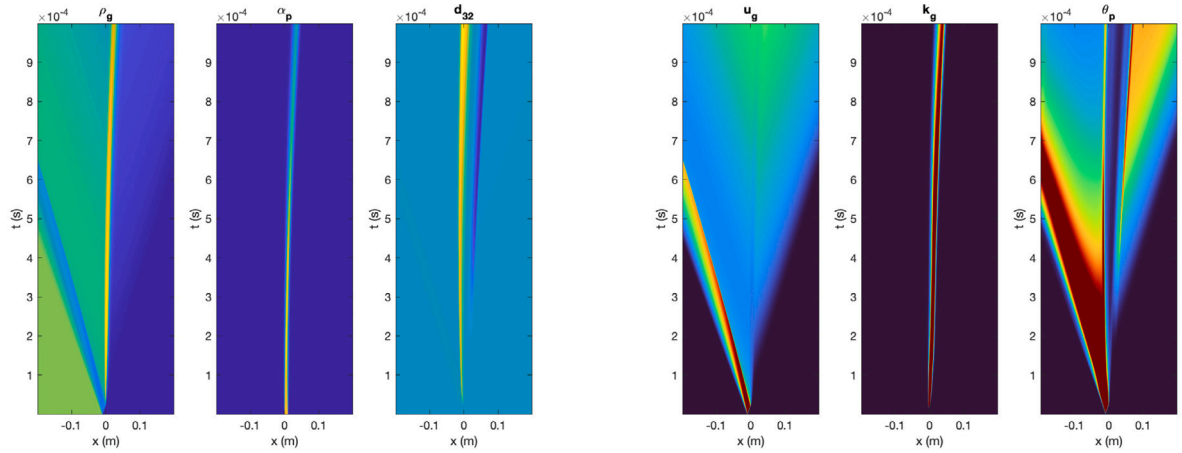


Fig. 18. Space–time contour plots for $\alpha_p = 0.1$. Left: gas density ρ_f, particle volume fraction α_p, mean diameter d_{32}. Right: gas velocity u_f, PTKE k_f, granular temperature Θ_p. For this moderately dense case, significant size segregation is present. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

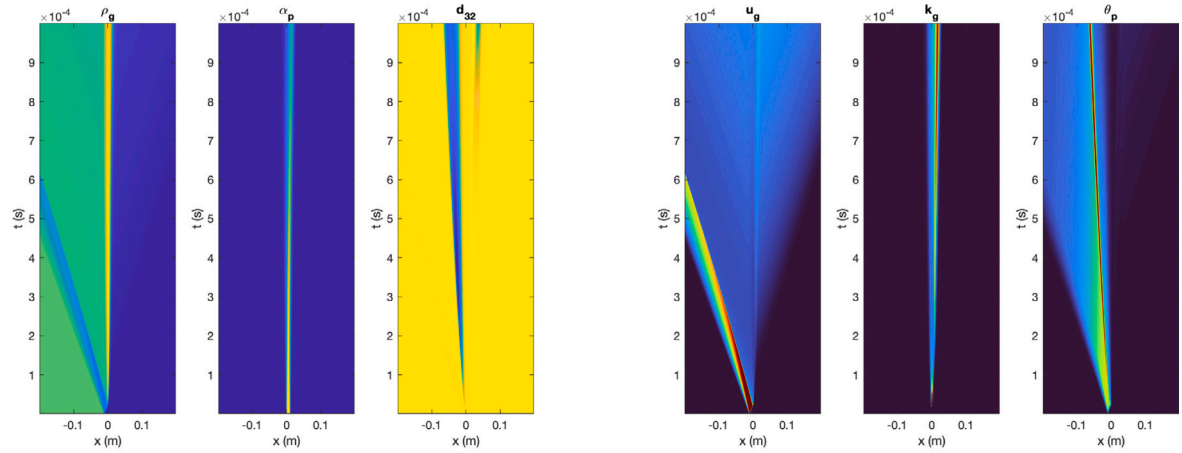


Fig. 19. Space–time contour plots for $\alpha_p = 0.6$. Left: gas density ρ_f, particle volume fraction α_p, mean diameter d_{32}. Right: gas velocity u_f, PTKE k_f, granular temperature Θ_p. For this very dense case, size segregation is present, but now also on the upstream side of the curtain due to small particles rebounding after collisions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

values for P_{pfp}. Because the particles have different sizes, the drag force will be different for each size, causing them to accelerate at different rates. However, the initial particle curtain is dense enough that particle–particle collisions are important, which causes the particle velocities to approach the mean value $\mathbf{u}_p$.

Eventually, the particle volume fraction drops as the curtain spreads and collisions become less important. The downstream particle size distribution (i.e., its mean and variance) will be determined by the differences in the spreading rate for particles of different sizes. For example, smaller particles should have higher velocities than larger ones, causing them to segregate as time advances. In the example results shown here, the initial curtain volume fraction α_p is set at different values to exhibit the differences in behavior between dilute to very dense cases. It should be pointed out that given the value of d_{max} and the relatively large value of α_p, the number concentrations of the particles in the curtain are much too high to allow for Euler–Lagrange simulations with resolved collisions. In fact, the macroscale model need not be solved in a 3-D spatial domain (which would be required for microscale and Euler–Lagrange simulations). Instead, it is possible to use the 1-D two-fluid model, which makes the computational cost relatively small. The numerical methods used to solve the macroscale model are based on hyperbolic solvers for multiphase flows (Houim and Kuo, 2011; Houim and Oran, 2016; Boniou and Fox, 2023; Boniou et al., 2023).

The space–time plots presented in Fig. 18 for moderately dense and in Fig. 19 for very dense initial conditions show how the profiles (horizontal axis) of the gas- and particle-phase variables evolve with time (vertical axis). In both cases, the shock is strongly reflected backwards from the curtain (see profiles of ρ_f and u_f). The particle volume fraction α_p moves to the right due to momentum exchange with the gas phase. As expected due to the difference in mass loading, the change in position is larger for the moderately dense case in Fig. 18. The evolution of the mean particle diameter d_{32} is different between the two cases. For the moderately dense case, the smaller particles (blue region) move further downstream than the larger particles (yellow region) due to drag-induced size segregation. In contrast, for the very dense case the smaller particles also move upstream. This is because the particle–particle collisions are very significant compared to drag and the smaller particles have higher granular temperature. The granular pressure of the smaller particles relative to the larger particles causes the upstream segregation.

The polydisperse macroscale model provides information concerning the size-conditioned variables, as well as the local particle size distribution (Fox et al., 2024). In Fig. 20, the GQMOM abscissae are plotted at the end of the simulation for the dilute (left) and moderately dense (right) cases. In each subplot, there are five curves. Four of them are for the abscissae ($N = 4$) and the green curve is for the mean diameter d_{32}. The values to the left and right are determined

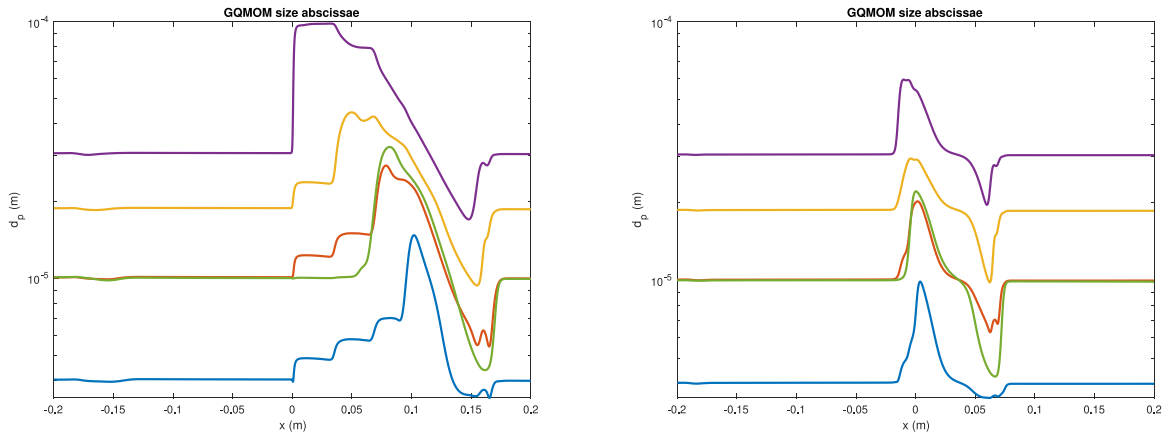


Fig. 20. Size abscissae and d_{32} (green) at $t = 10^{-3}$ s. Left: $\alpha_p = 0.001$. Right: $\alpha_p = 0.1$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

by the initial distribution in Fig. 17. Note that in these regions the GQMOM weights are nearly zero because there are no particles. As seen for the initial conditions, the values of the four abscissae provide information concerning the local particle size distribution, while d_{32} gives its average value. For example, with $\alpha_p = 0.001$ the mean value to the left of $x = 0$ is nearly constant but the abscissae jump up. This indicates that the local distribution is no longer a beta distribution but is skewed towards larger values while keeping the same average. Physically, this is a result of drag-induced size segregation. For both cases shown in Fig. 20, d_{32} varies significantly, decreasing left to right due to size segregation.

In summary, for this example with complex flow physics, much can be learned from the 1-D macroscale model due to its computational efficiency relative to the micro/mesoscale models. Nevertheless, the accuracy of the macroscale predictions depends on the closures developed for the mesoscale model and validated using microscale PR-DNS data. For example, the polydisperse collision model and the fluid-drag coefficient must be validated for high-speed flows, while for liquid–particle flows a validated model for the pfp stresses is required. In any case, the closures developed for the moments found from the kinetic equation must always result in a well-posed macroscale model. Otherwise, the numerical solutions found with the macroscale model will not be physical, and certainly will not reproduce the flow statistics found from solving the mesoscale model.

7. Conclusions

The kinetic-based approach for modeling polydisperse multiphase flows introduces an intermediate step between PR-DNS at the microscale and the Eulerian two-fluid model at the macroscale. In this work, the relationships between the models at the three different scales have been discussed. From a physical-modeling standpoint, the transition from the microscale to the mesoscale model is the critical step. In general, the most difficult challenge for this transition is the successful modeling of terms resulting from interactions between particles, including fluid stresses transmitted via the continuous phase. For example, in recent years, much progress has been made on this topic in the context of mesoscale Euler–Lagrange simulations where the location and velocity of each particle is known (see, e.g., Siddani and Balachandar (2023)). A valuable future research contribution would be to translate these advances into a model for the pfp stresses appearing in the mesoscale kinetic equation in which the particle locations are unknown.

In the literature, much of the recent focus in the field of polydisperse multiphase flows has been on microscale PR-DNS of various multiphase systems. For example, excellent progress has been made in PR-DNS of non-spherical solid particles, droplets and bubbles with and/or without breakage and coalescence. However, most of these studies focus on the microscale physics as opposed to collecting the data required to postulate and validate mesoscale closures, especially in denser systems where particle–particle interactions are strong. This, of course, is as expected because one must first understand the physical mechanisms before conceptualizing a kinetic-based closure for the process. In the future, as the PR-DNS databases expand, it will be important to design PR-DNS 'experiments' that measure the quantities appearing in the kinetic models, which are not always obvious nor straightforward to calculate.

As a specific example, one can point to the pfp stresses appearing in the particle-phase momentum balance as a spatial flux (or pressure tensor). Recent work has demonstrated the importance of such stresses in well-posed macroscale models (Fox et al., 2020). In the literature, most of the mesoscale modeling based on PR-DNS has focused on the source terms (e.g., drag, lift, heat transfer, etc.) that can be extracted from spatially homogeneous PR-DNS simulations, but do not change the hyperbolicity of the macroscale model. In comparison, extracting information concerning spatial fluxes (or stresses) from PR-DNS is more difficult (Wang et al., 2021), as is designing a flow configuration that isolates such terms. For example, the pfp stresses affect the wavelength of kinematic waves as shown in Fig. 12. But to obtain this information from a PR-DNS, the simulation domain must be large enough to allow for variations in the volume-fraction field at the meso/macroscale. This is currently possible only when the material density ratio $\mathcal{Z}$ is sufficiently large (e.g., liquid–solid flows). Thus, additional effort is needed to find other methods to extract the pfp stresses from PR-DNS results.

As mentioned above, the most difficult source terms to close at the mesoscale are those resulting from particle–particle interactions. For example, even when using an Euler–Lagrange approach, modeling collisions when $\mathcal{Z}$ is sufficiently large is not straightforward because of the presence of the fluid. The same is true for aggregation and/or coalescence events for which the kernels appearing in the mesoscale model must be parameterized in terms of the resolved fluid-phase variables (e.g., shear stress, surface tension, etc.). Even the 'simple' phenomena of drop breakup in turbulence is challenging to parameterize at the mesoscale in terms of the classical breakage frequency and child-size distribution. In particular, the microscale physics that determine the child-size distribution are not always consistent with a binary-breakup model (Farsoiya et al., 2023). Similar challenges also arise in other systems such as bag breakup in sprays. It is for this reason that the formulation of the mesoscale model (either Euler–Lagrange or kinetic-based) plays a crucial role in determining the predictive ability of the meso/macroscale models.

Assuming that the solution to the mesoscale model is reasonably accurate, the passage to a macroscale model can be envisioned. Fortunately, this passage involves mathematical (as opposed to physical)

closures. (Recall that the mesoscale model resolves the fluid-phase physics and, therefore, no turbulence model is used for length scales above the particle size.) For this to be possible, the kinetic equation for the particle velocity distribution must have source terms such as collisions or fluid drag that control the shape of the velocity distribution function. In this work, we have assumed that the hydrodynamic approximation (i.e., mass, momentum and kinetic energy) for the particle phase is sufficient at the macroscale. This approximation can be extended by including higher-order velocity moments; however, unless the Stokes number is small enough to control the velocity distribution function, eventually it will be necessary to solve the mesoscale model using PIC or DSMC. In cases where the (extended) hydrodynamic model is accurate, the macroscale model for monodisperse particles takes the form of an Eulerian two-fluid model. By accounting for added mass and the pfp stresses, this Eulerian model is well-posed for arbitrary $\mathcal{Z}$ (Fox et al., 2020).

For polydisperse particles, the passage from the mesoscale to the macroscale model leads to unclosed equations for a set of moments representing the particle-size distribution function and the correlation between the particle size and velocity (Fox et al., 2024). The mathematical challenge is then to close accurately the moment balance equations using only a small set of moments. In the literature, this is often accomplished using a presumed distribution function (e.g., joint Gaussian, etc.) or sectional methods (i.e., dividing the size variable into a few fixed sizes). In this work, quadrature-based moment methods (QBMM) have been put forward as a more accurate alternative for many applications, especially those involving growth, coalescence/aggregation and/or breakage (Marchisio and Fox, 2013). However, as with the velocity distribution discussed above, it is important to verify that the chosen moments used for closure at the macroscale are sufficient to reproduce the mesoscale solution to the population balance equation for the particle-size distribution function. Fortunately, for particle systems with small Stokes number (i.e., flow followers), such verification studies for many different types of coalescence/aggregation/breakage kernels have already been carried out using QBMM (Marchisio and Fox, 2013). The same is true for particles with more properties in addition to size (e.g., surface area, temperature, chemical composition, etc.). As described in Section 5.4, QBMM also provides an efficient representation of the size–velocity correlation in complex flows such as sedimentation (Shajahan and Breugem, 2023) and shock-particle-curtain interactions (Capecelatro and Wagner, 2024).

Finally, in this work, the focus has been on disperse multiphase flow modeling. However, in many important applications such as liquid-jet break up to form droplets (Vu et al., 2023) or ocean waves breaking to produces bubbles and sprays (Deike, 2022), the disperse phase is created from a segregated multiphase flow. To model such flows across both the segregated and disperse regimes, it will be necessary to couple the Eulerian disperse-flow models described in the work with Eulerian segregated-flow models like the one proposed in Furfaro and Saurel (2015). This coupling will require novel ideas, such as using geometrical moments (Essadki et al., 2019), to ensure that the resulting two-fluid models are consistent and well-posed for all possible regimes occurring in the flow domain. Moreover, advances in the numerical solvers to ensure that they are robust and accurate across the interfaces separating the segregated and disperse regime will also be required.

CRediT authorship contribution statement

Rodney O. Fox: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

Due to the huge number of high-quality scientific publications dealing with the topics covered in this work, I was unable to cite many significant contributions by numerous colleagues working in polydisperse multiphase flows. Despite this obvious shortcoming, I hope to have been able to have transmitted the general philosophy of kinetic-based modeling approaches to readers new to this field, and possibly to have inspired future research to bridge the scales identified in Fig. 2 and in segregated/disperse flows.

When preparing this work, I was supported by the Fulbright–Tocqueville Distinguished Chair Award, France and the Jean D'Alembert Senior Professor Chair at the Université de Paris-Saclay, France. The Government of the United States or any agency representing it has not endorsed the conclusions or approved the contents of this publication.

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